

23. April 2026

Figures

Collection

started May 2024

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1 Setups and devices

Fig. zz044: THz setup

- Origin: Paper IRMMW and THz

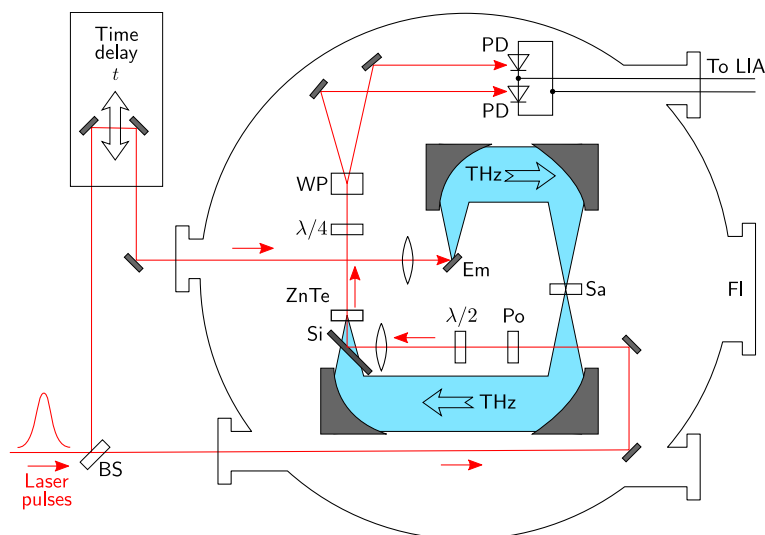


Fig. 1.1: File: zz044001.eps. THz setup

Fig. zz030: Various molecular and polymeric semiconductors:

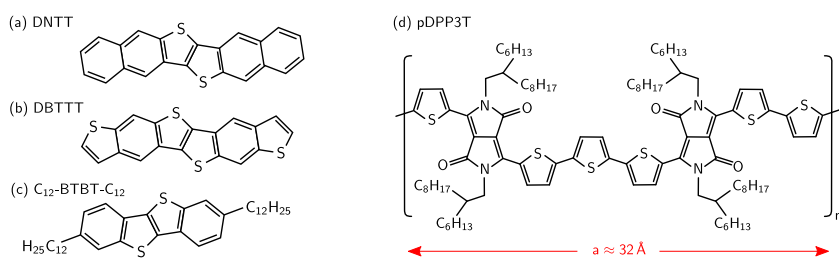


Fig. 1.2: File: zz030002.eps. Molecular and polymeric semiconductors.

Fig. zz005 Schematics of transport and THz electromodulation: spectroscopy

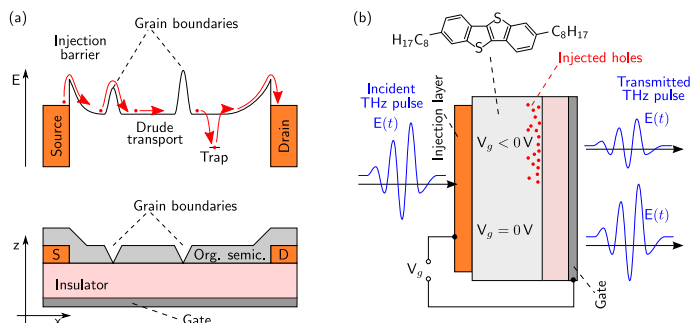


Fig. 1.3: File: zz005001.eps. Schematics of transport and of THz electromodulation spectroscopy.

Fig. zz043: Structure of THz device and photo

- Origin: Paper IRMMW and THz

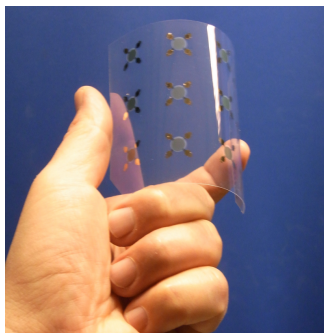
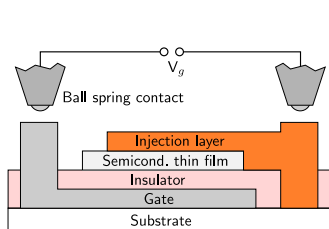


Fig. 1.4: File: zz043002.eps. Structure of THz device and photo.

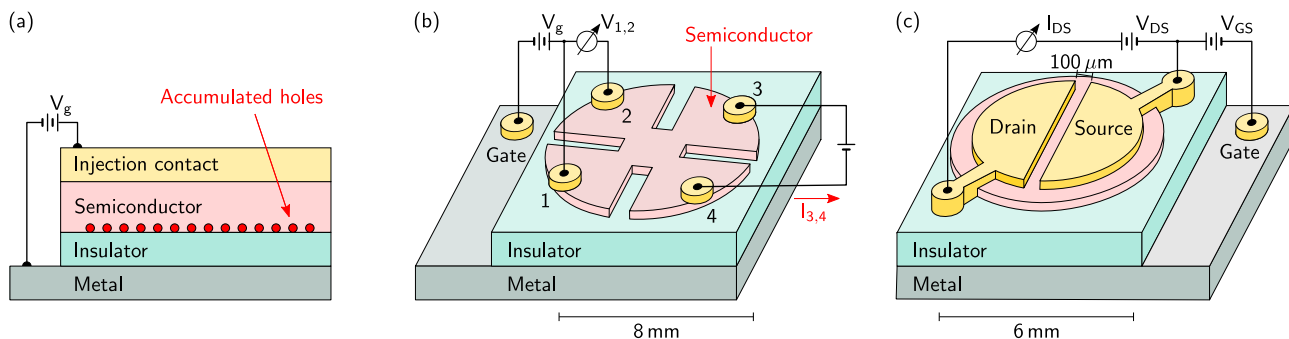


Fig. 1.5: File: zz096009.eps. (a) Cross-section of an organic field-effect device. (b) device for van-der Pauw characterization, and (c) device for conventional electronic characterization, which can be also used for THz spectroscopy.

Fig. zz096: Various device structures

Fig. zz111: CVD layout

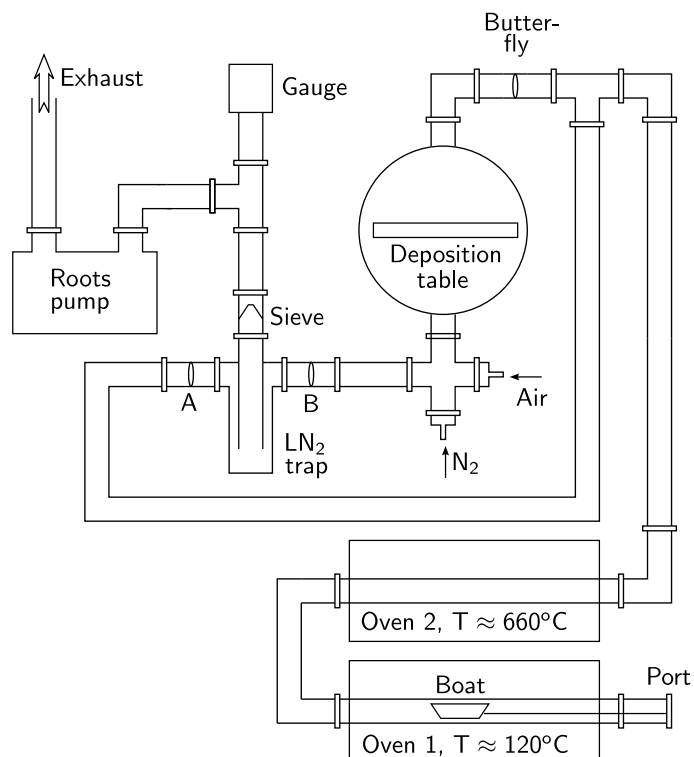


Fig. 1.6: File: zz111001.eps. CVD layout.

Fig. zz155: Clausis-Clapeyron law for parylene N

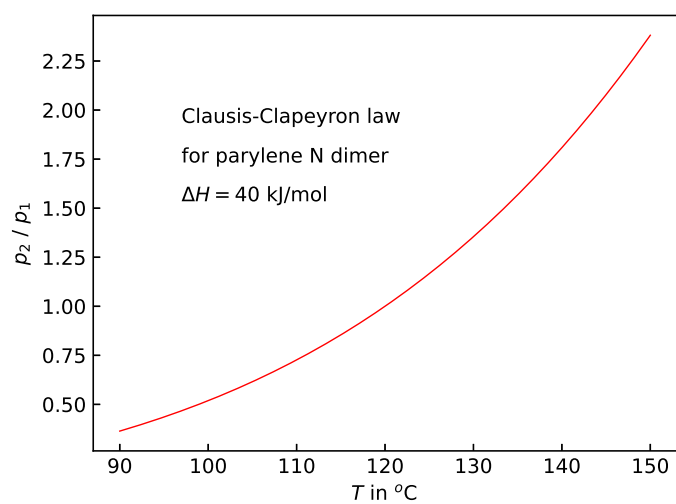


Fig. 1.7: File: zz155.eps. Clausius-Clapeyron law for parylene N.

2 Terahertz

Fig. zz077: Properties of the THz pulses

- Origin: Paper IRMMW and THz
- python: zz077005.py

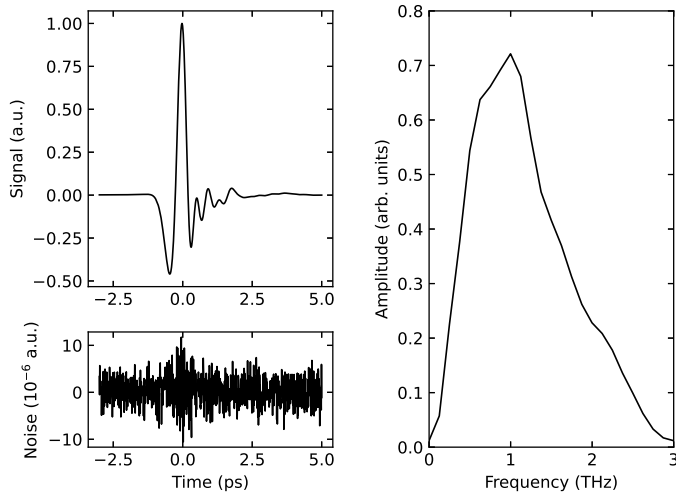


Fig. 2.1: File: zz077005.eps.

Fig. zz117: Amperian loop / differential signals

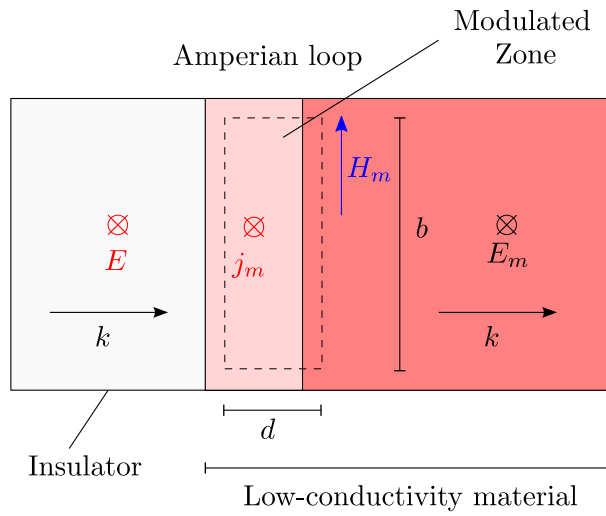


Fig. 2.2: File: zz117001.eps. Explaining differential THz signals with an Amperian loop

3 Inorganic Semiconductors

Fig. zz000:

- Origin: none

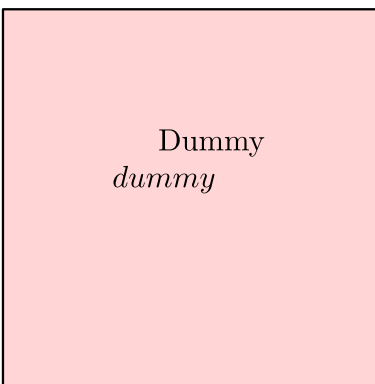


Fig. 3.1: File: zz000000.eps. Dummy.

4 Molecular semiconductors

Fig. zz002: Popular molecular semiconductors

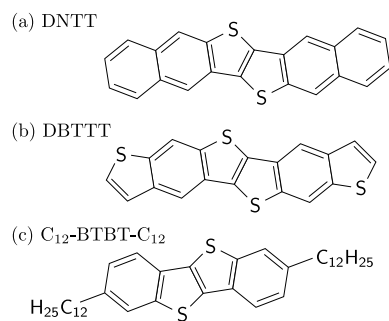


Fig. 4.1: File: zz002001.eps. Popular molecular semiconductors.

Fig. zz003: Bandwidth narrowing due to polarons

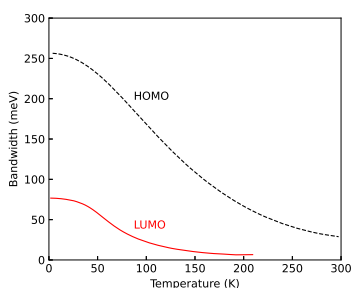


Fig. 4.2: File: zz003001.eps. Bandwidth narrowing due to polarons as calculated by Hannewald.

Fig. zz004: Dynamic localization

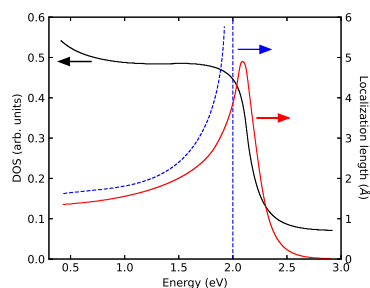


Fig. 4.3: File: zz004003.eps. Dynamic localization as calculated by Fratini 2016.

Fig. zz005:

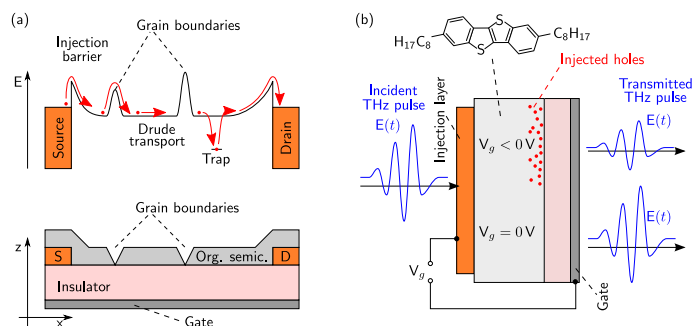


Fig. 4.4: File: zz005001.eps.

Fig. zz006: C8-BTBT, recorded on ms range

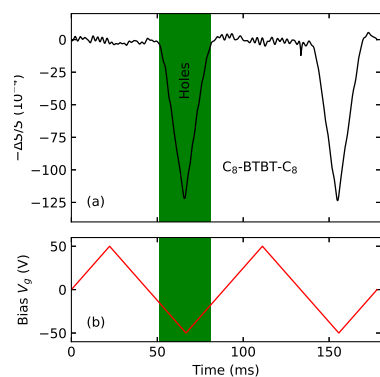


Fig. 4.5: File: zz006001.eps. C8-BTBT on ms range.

Fig. zz129: C8-BTBT on PaN + PS, ms range

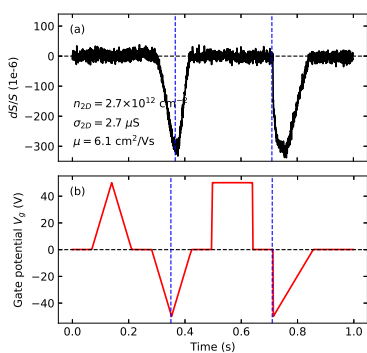


Fig. 4.6: File: zz129001.eps. Transient on ms scale recorded on C₈-BTBT-C₈ deposited on sapphire + PaN + PS. Data file: 2025080808.dat.

Fig. zz019:

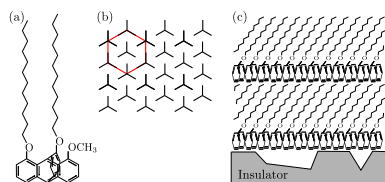


Fig. 4.7: File: zz0019001.eps. Triptycene-C12.

Fig. zz021: Dph-BTBT on hBN

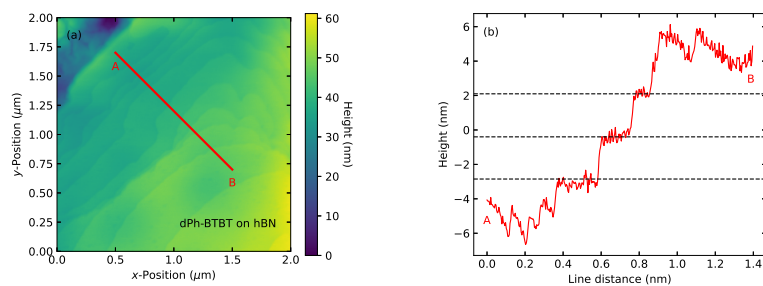


Fig. 4.8: File: zz0021001.eps. Dph-BTBT on hBN.

Fig. zz023001: DNTT

- Description:
 - DNTT
 - THz ms-range
 - by: P.R.
- Sample:
 - sample number: N01-02
 - cooperation: Nickel
- Original data:
 - Folder: Q:\...AllMeasuredData\2023\20230123\N01-02 DNTT ambipolar
 - File: 20230123000_N01-02 DNTT ambipolar_interesting modulation.ave

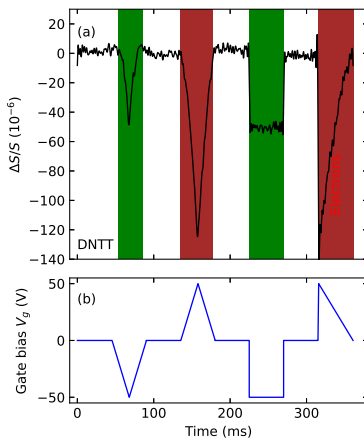


Fig. 4.9: File: zz023001.eps. DNTT ambipolar THz absorption in ms-domain.

Fig. zz008001: C12-BTBT time domain

- Description:
 - C12-BBT
 - THz ps-range
- related Fig. 4.11:
 - ?
- Original data:
 - ?

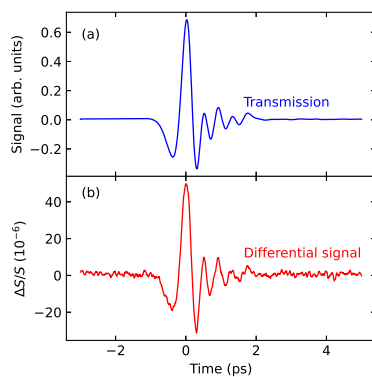


Fig. 4.10: File: zz008001.eps. C12-BTBT in time domain.

Fig. zz009001: C12-BTBT in frequency-omain

- Description:
 - C12-BBT
 - THz Fourier analysis
- related Fig. 4.10:
 - ?
- Original data:
 - ?

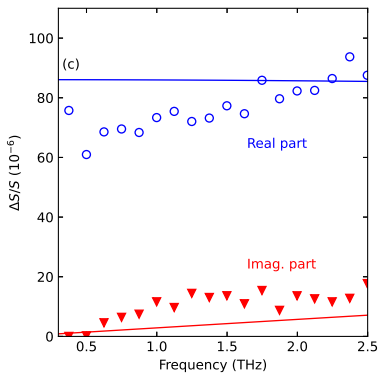


Fig. 4.11: File: zz009001.eps. C12-BTBT in time domain.

Fig. zz076001: C8-BTBT-C8 ms-scale

- Description:
 - C8-BTBT-C8
 - original figure: JIRMMW

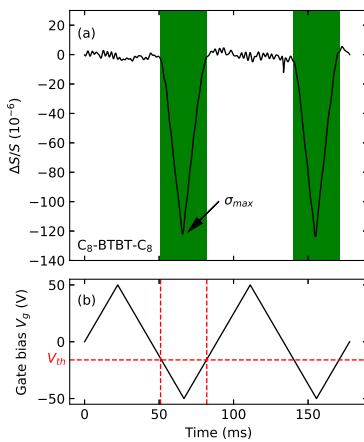


Fig. 4.12: File: zz076001.eps. C8-BTBT-C8 in ms time domain.

Fig. zz078001: Transfer technology

- Description:
 - Polyacrylic acid
 - Kalkan et al. [1]

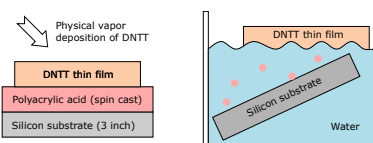


Fig. 4.13: File: zz078001.eps. Film transfer.

Fig. zz083001: DBTTT

- Purpose: Ioffe-Regel criterion
- Measurement: Temperature dependence of mobility
- Paper: Molecular semiconductors and the Ioffe-Regel criterion: A terahertz study on band transport in DBTTT, Fig. 5

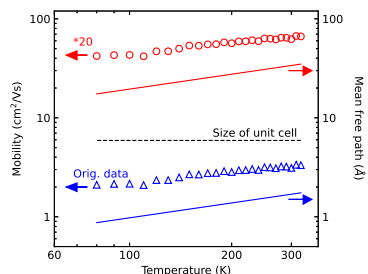


Fig. 4.14: File: zz083001.eps. Temperature dependent measurements on DBTTT

Fig. zz083001: DBTTT

- Purpose: Ioffe-Regel criterion
- Measurement: Temperature dependence of mobility
- Paper: Molecular semiconductors and the Ioffe-Regel criterion: A terahertz study on band transport in DBTTT, Fig. 5

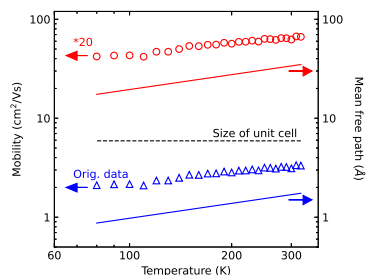


Fig. 4.15: File: zz083001.eps. Temperature dependent measurements on DBTTT

Fig. zz101001: Binding energy of traps

- Paper Ke-Riederer2022a.pdf

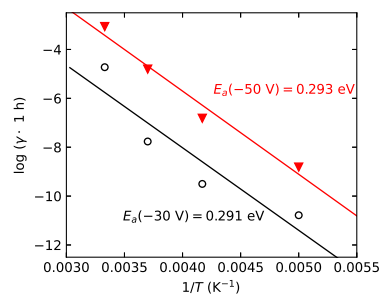


Fig. 4.16: File: zz101001.eps. Binding energy of traps.

Fig. zz148: Relative differential THz transmission of the ms-scale

- ms-scan of Parkar
- data: 20260205009.dat
- sample: 20253010_10 (PaN+PS) ???, which material
- python: zz148

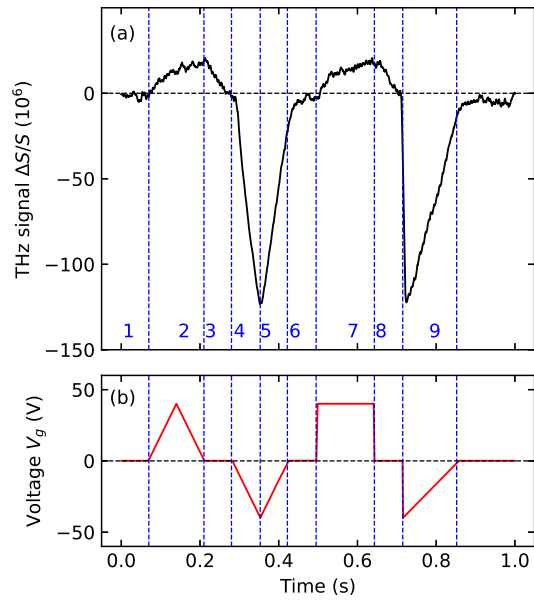


Fig. 4.17: File: zz148.eps. Relative differential THz transmission on an unknown sample.

5 Polymeric semiconductors

Fig. zz050: Structures of polymers

- References:
 - Kayunkid 2014 [2]
 - Alesadi 2021 [3]

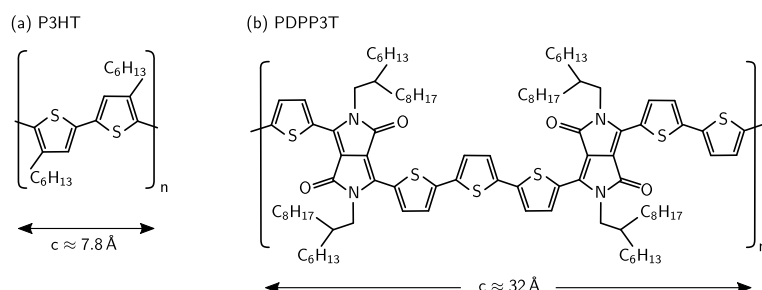


Fig. 5.1: File: zz050004.eps. Structure of P3HT and of pDPP3T. Also shown are the sizes of the unit cells $c = 7.8 \text{ \AA}$ and $c = 32 \text{ \AA}$ of P3HT and PDPP3T, respectively. [2, 3]

Fig. zz097: Structures of polymers

- References:
 - Kayunkid 2014 [2]
 - Alesadi 2021 [3]
 - Alberga 2015 [4]

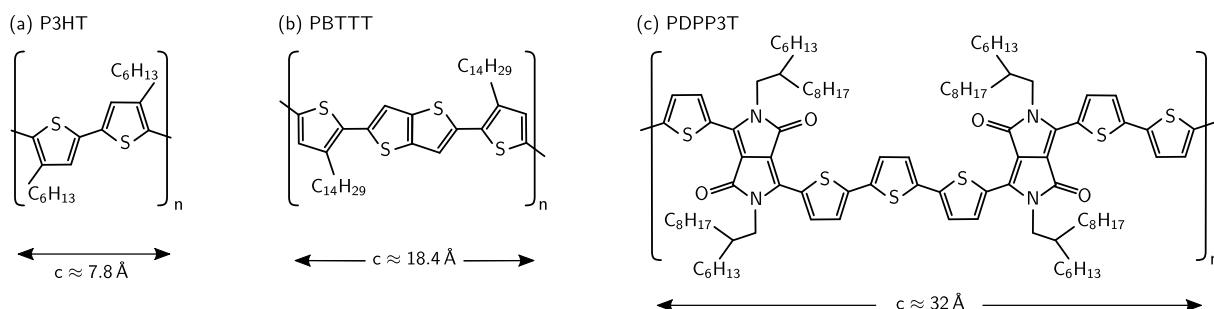


Fig. 5.2: Polymeric semiconductors, which are most relevant to the project. Structure of regio-regular P3HT (a), of PBTTT, and of the dono-acceptor copolymer PDPP3T (c). Also shown are the sizes of the unit cells along the polymer backbones of $c \approx 7.8 \text{ \AA}$, $c \approx 18.4 \text{ \AA}$, and $c \approx 32 \text{ \AA}$ for P3HT, PBTTT, and PDPP3T, respectively [2–4].

Fig. zz059: First monolayers of aggregates

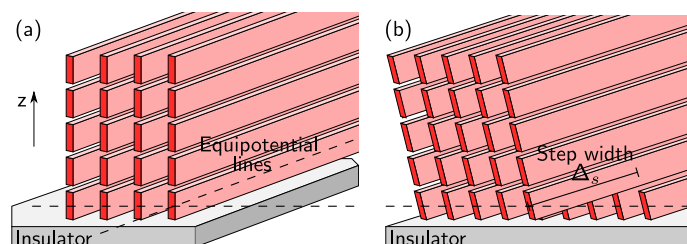


Fig. 5.3: File: zz059032.eps. Orientation of the polymer chains (red blocks) within an aggregate. a) Idealized aggregate with a planar first monolayer. All molecules of this monolayer have the same potential. b) More realistic aggregate with monolayer steps at the interface with the insulator. Here, only part of the molecules are intersected by the equipotential surface.

Fig. zz064: Nanostructure of thin films

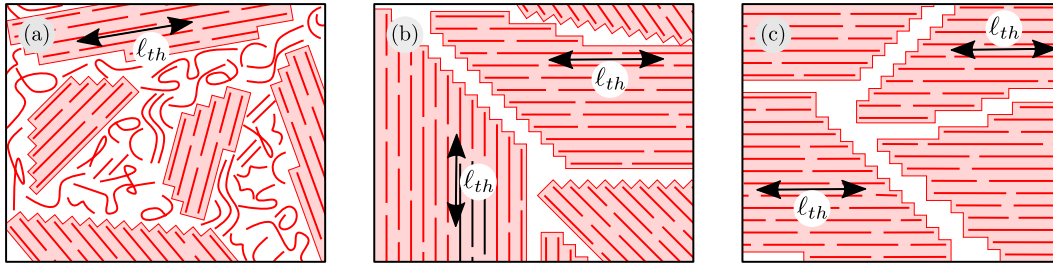


Fig. 5.4: File: zz064019.eps. Schematics of the nanostructure of semiconducting polymer thin-films. (a) Disordered aggregates and amorphous regions. (b) Crystalline thin-film with arbitrary orientation of the aggregates. (c) Oriented nanocrystals. The backbones of the polymer chains are depicted by solid red lines, with distortions of the π -conjugation illustrated by the interruptions. The thermal mean free path is indicated by ℓ_{th} .

Fig. zz007001: pDPP3T on ms-range

- Description:
 - pDPP3T on ms range
 - THz ms-range
 - by: P.R.
- Sample:
 - sample number: ?
 - cooperation: Weitz
- Original data:
 - see folder of paper

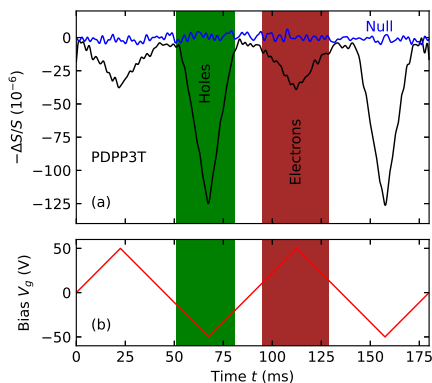


Fig. 5.5: File: zz007001.eps. Terahertz transmission through PDPP3T [5]. (a) Relative differential transmission $\Delta S/S$ in dependence on the applied bias shown in (b). At $V_g < 0$ mobile holes are injected into the device and at $V_g > 0$ mobile electrons are injected. The transmission reduces when electrons or holes are injected into the device.

Fig. zz031001: pDPP3T rotation

- Description:
 - pDPP3T on ms range
 - THz ms-range
 - by: P.R.
- Sample:
 - sample number: ?
 - cooperation: Weitz

- Original data:
 - see folder of paper

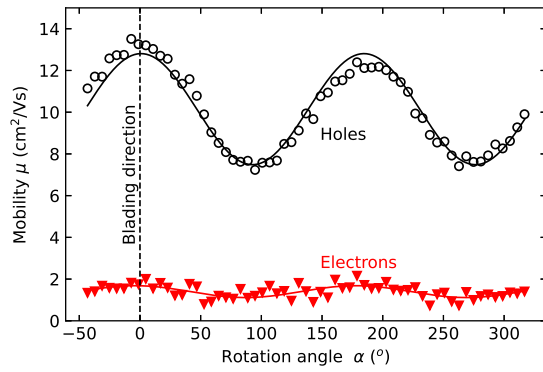


Fig. 5.6: Electron and hole mobility in PDPP3T. Dependence of the mobilities on the angle α between the polarization of the THz beam and the preferred orientation of the PDPP3T crystals [5].

Fig. zz032001: PDPP3T AFM analysis

- Description:
 - AFM of pDPP3T
 - by: P.R.
- Original data:
 - see folder of paper

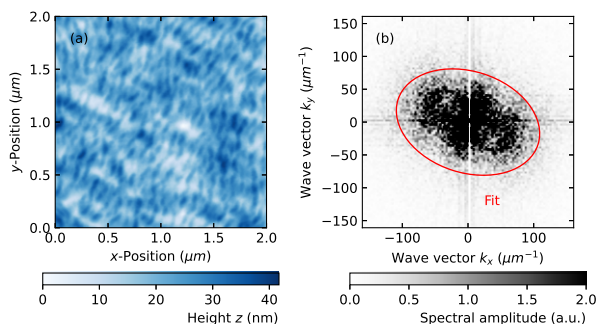


Fig. 5.7: File: zz032002.eps. (a) AFM image of a PDPP3T thin-film fabricated by blade coating [5]. (b) Fourier transform of (a). The ellipse shows a fit of the data using the dependence of the hole mobility on orientation as illustrated in Fig. 5.6.

Fig. zz085001: PDPP3T AFM analysis

- Description:
 - AFM of pDPP3T, not aligned
 - by: P.R.
- Original data:
 - see folder of paper

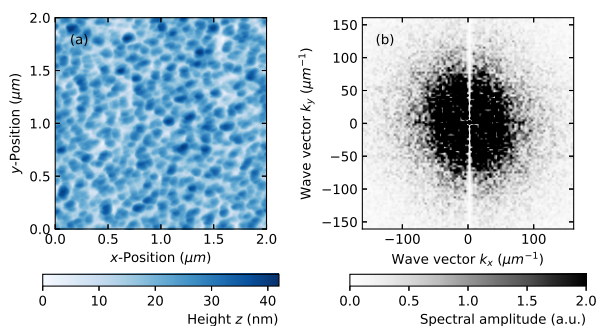


Fig. 5.8: File: zz085002.eps. (a) AFM image of an unaligned PDPP3T thin-film fabricated by blade coating [5]. (b) Fourier transform of (a).

Fig. zz033001: pDPP3T modulation decay

- Description:
 - pDPP3T on ms range
 - THz decay on second range
 - by: P.R.
- Original data:
 - see folder of paper

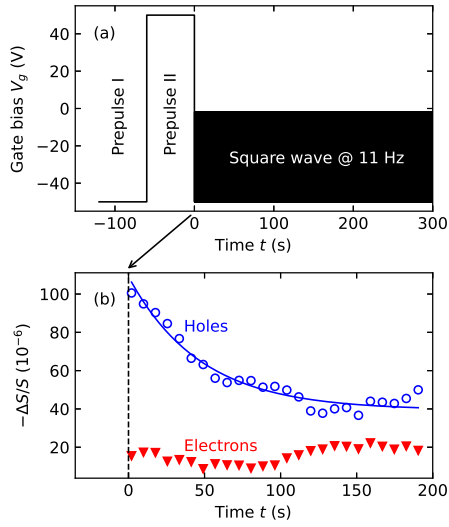


Fig. 5.9: File: zz033001.eps. Measurement of the decay of the density of mobile charge carriers [5]. (a) Applied bias. The positive and the negative pre-pulse at $t < 0$ s remove trapped carriers from the device and bring it into a well-defined state. The square wave signal after $t = 0$ s is used to measure the relative differential signal $\Delta S/S$ of the THz transmission (b) Evolution of $\Delta S/S$ after $t = 0$ s. When probing the properties of holes, a significant decay of $-\Delta S/S$ is observed on the time scale of 50 s. For electrons no reliable data were obtained, presumably, because this signal decays on shorter time scales.

Fig. zz034001: Band structure and trapping

- Description:
 - from pDPP3T paper

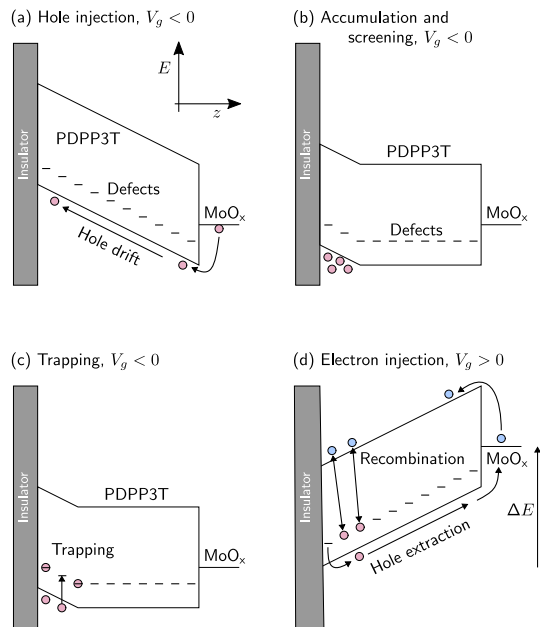


Fig. 5.10: File: zz034001.eps. Band structure from pD-PP3T paper.

Fig. zz035001: Temperature dependence of THz transmission through pDPP3T

- Description:
 - from pDPP3T paper

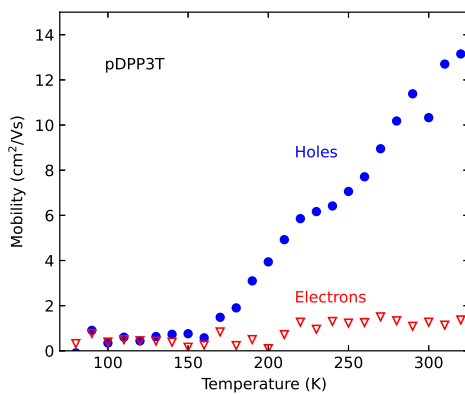


Fig. 5.11: File: zz035001.eps. Temperature dependence of THz transmission through pDPP3T.

Fig. zz052: Charge modulation data of Enengl

- References:
 - Enengl, 2016 [6]

Fig. zz057: Trapping mechanisms

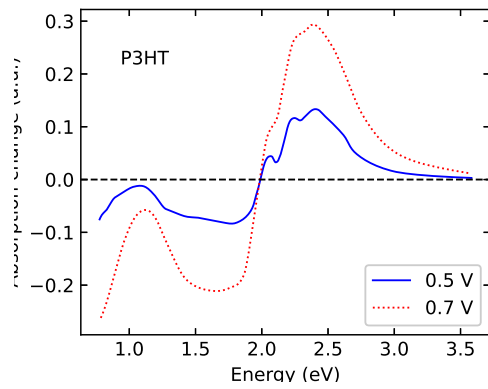


Fig. 5.12: File: zz052001.eps. Charge modulation spectra recorded on P3HT. Data from Enengl et al. [6].

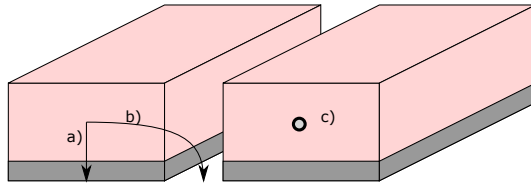


Fig. 5.13: File: zz057001.eps. Traps and trapping mechanisms.

Fig. zz058: Disorder in the first ML

Used values:

- monomer unit cell $V = 1 \times 1 \times 1 \text{ nm}^3$
- Ioffe Regel criterion $\ell_{th} = 10 a$
- if 19/20 carriers are trapped the situation similar to Fig. 5.14

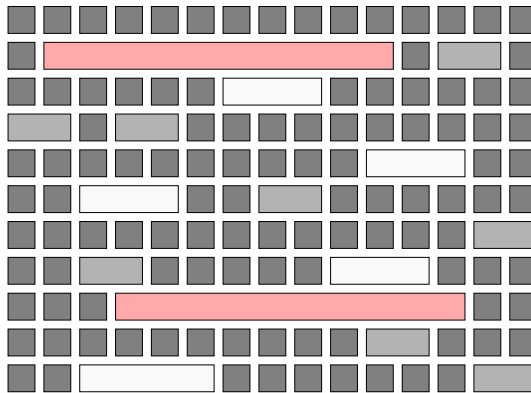


Fig. 5.14: File: zz058001.eps.

Fig. zz061: Model for trap distribution and filling

Fig. zz063: MC calculations by Noriega

Fig. zz065: Calculations of the DOS by Rivnay

Source: Rivnay et al. [8]

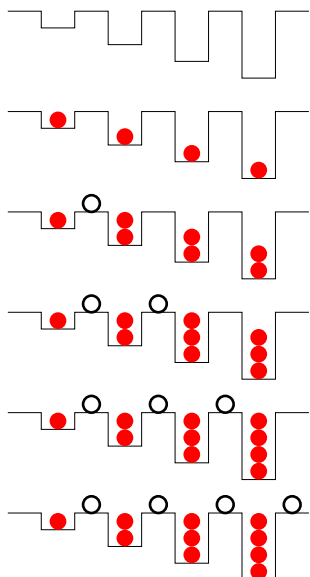


Fig. 5.15: File: zz061001.eps. Model for trap distribution and filling

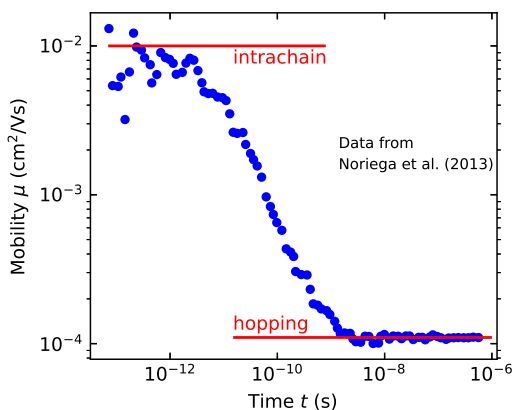


Fig. 5.16: File: zz063001.eps. MC simulations by Noriega et al. [7].

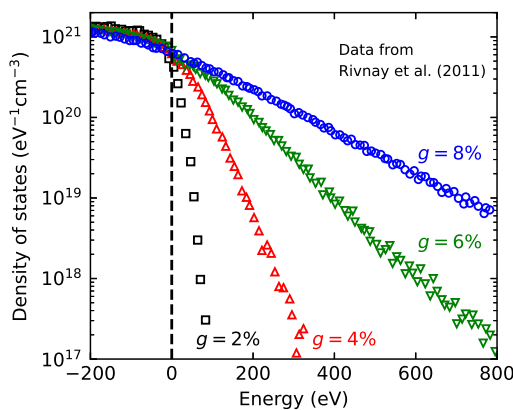


Fig. 5.17: File: zz065001.eps. DOS calculated by Rivnay et al. [9].

Fig. zz067: Absorption spectra of P3Ht thin films

Source:

- thin film of P3Ht
- data: 20240528
- description: treatment of the P3Ht solution in acetone vapor

Fig. zz068: Absorption spectra of P3HT thin films

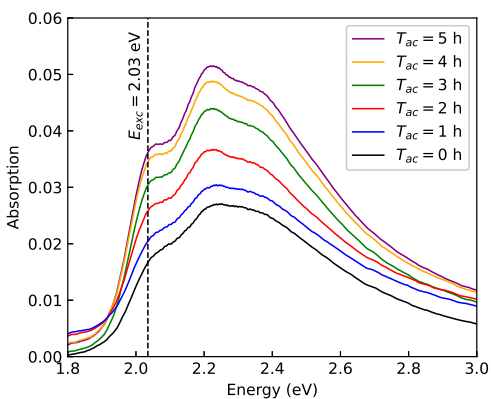


Fig. 5.18: File: zz067001.eps. Absorption spectra of P3HT. The parameter T_{ac} is the time the liquid solution was under acetone vapor.

Source:

- thin films of P3Ht
- data: 20240619
- description: treatment of the P3Ht solution in acetone vapor

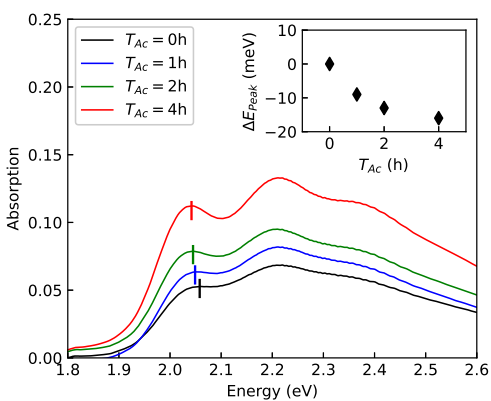


Fig. 5.19: File: zz068001.eps. Absorption spectra of P3HT. The parameter T_{Ac} is the time the liquid solution was under acetone vapor.

Fig. zz071: Crystal structure of P3HT

Source:

- Z. Wu 2020, [10]
- lattice constants: $a = 15.7 \text{ \AA}$, $b = 7.7 \text{ \AA}$, and $c = 7.6 \text{ \AA}$

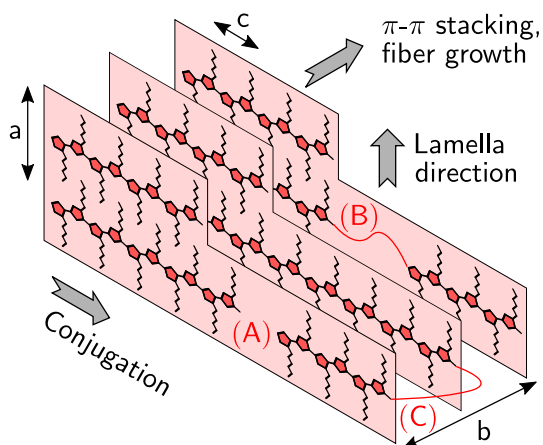


Fig. 5.20: File: zz071007.eps. Crystal structure of P3HT. Inspired by Ref. [10], which provides lattice constants of $a = 15.7 \text{ \AA}$, $b = 7.7 \text{ \AA}$, and $c = 7.6 \text{ \AA}$. Defects such as disruptions of the π -conjugation, disordered links, and hairpin bending are indicated by A, B, and C, respectively.

ppp ppp ppp ppp ppp ppp llll ppp ppp ppp ppp ppp ppp ppp ppp ppp llll ppp ppp ppp ppp ppp ppp
ppp ppp ppp ppp llll ppp ppp ppp ppp ppp ppp ppp ppp ppp ppp llll ppp ppp ppp ppp

Fig. zz075: Thin-films of P3HT after acetone treatment

Source: AFM data

- 2024062001.spm, $T_{ac} = 0$ h
- 2024062002.spm, $T_{ac} = 1$ h
- 2024062403.spm, $T_{ac} = 4$ h

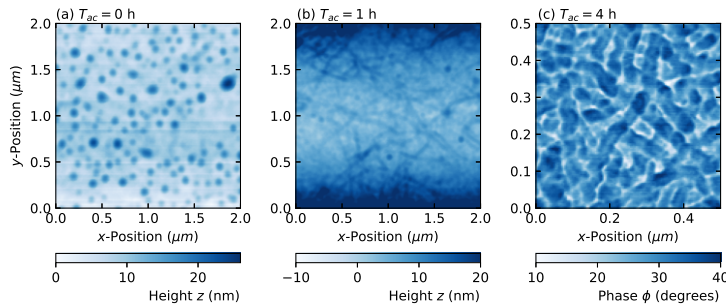


Fig. 5.21: File: zz075001.eps. AFM images recorded on spin-cast thin-films of P3HT. (a) Using a pristine 0.1 wt% solution in toluene. Data after exposing the solution for $T_{ac} = 1$ h and for $T_{ac} = 4$ h to an acetone atmosphere shown in (b) and (c), respectively. Mind the different dimensions of the images and that (c) shows phase data.

Fig. zz079: Coating by solution shearing

- alias blade coating
- see Shaw, [11]

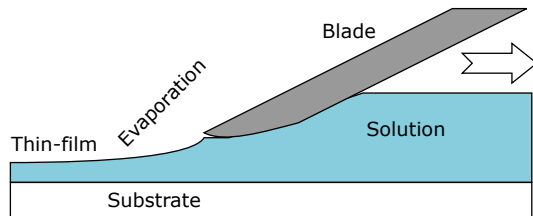


Fig. 5.22: File: zz079001.eps.

Fig. zz084: Device modulation

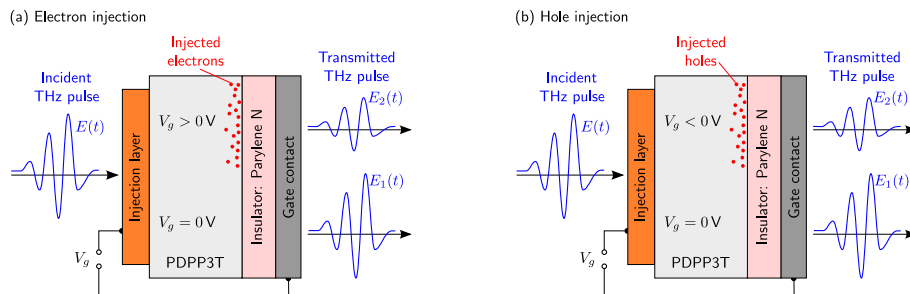


Fig. 5.23: File: zz084001.eps. Modulation of PDPP3T-devices.

Fig. zz086: Absorption spectra of P3Ht and PDPP3T in toluene

- P3Ht in toluene
- PDpp3T in toluene
- data from 20240704

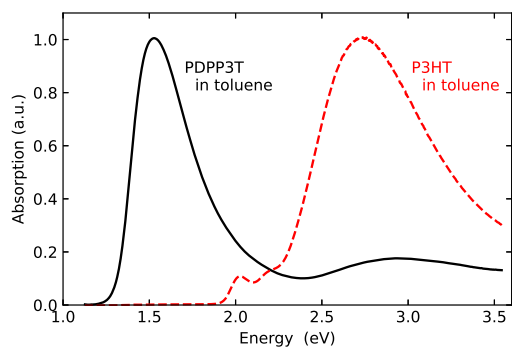


Fig. 5.24: File: zz086001.eps.

Fig. zz087: Orientation by imprint

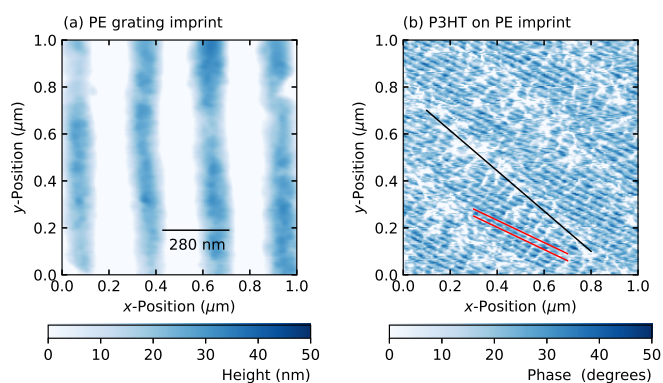


Fig. 5.25: File: zz087001.eps.

Fig. zz090: Schematic on polarization

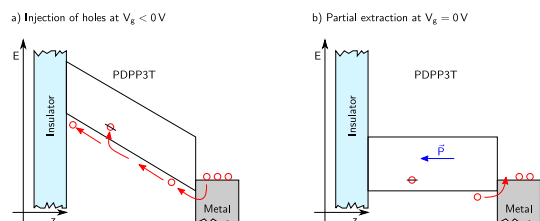
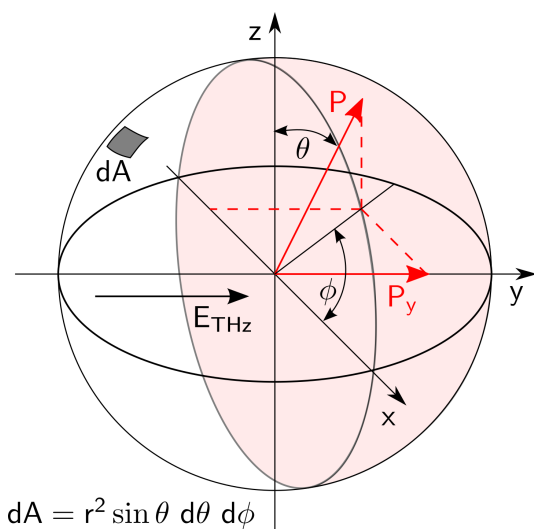


Fig. 5.26: File: zz090001.eps. Schematic on polarization.

Fig. zz099: Orientation of dipole moment



$$dA = r^2 \sin \theta \, d\theta \, d\phi$$

Fig. 5.27: File: zz099002.eps. Projection of the lightfield E_{THz} onto the orientation of the main dipole moment of the polymer P .

6 Calculations

Fig. zz091a: Localization-modified model after Mott

- description: Mott and Kaveh 1985
- AllMyPhys02.tex, section 39

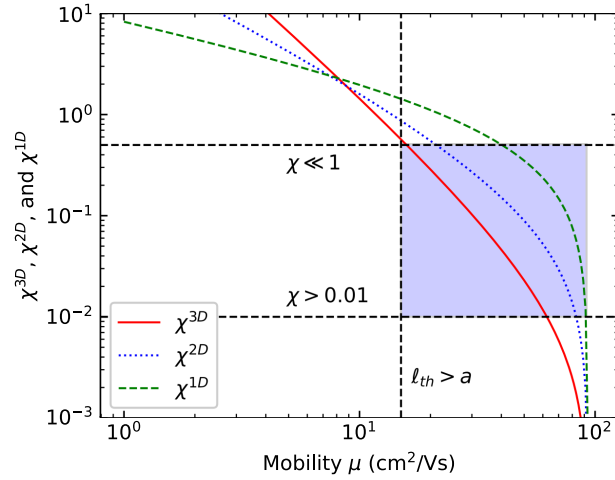


Fig. 6.1: File: zz091002a.eps. Correction factors of localization-modified models.

Fig. zz092a: Localization-modified model after Mott

- description: Mott and Kaveh 1985
- AllMyPhys02.tex, section 39

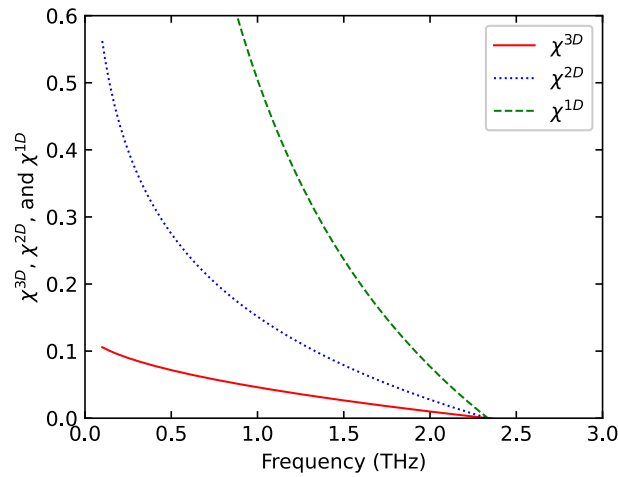


Fig. 6.2: File: zz092001a.eps. Correction factors of localization-modified models.

Fig. zz094: Comparison of models with data obtained on DBTTT

- supplementary of Riederer 2023 [12]

Fig. zz103: Trapping model 1

Fig. zz105: Trapping model 2

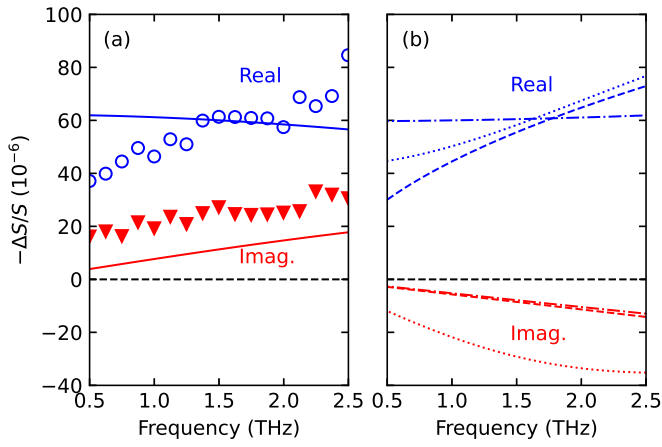


Fig. 6.3: File: zz094001.eps. Impact of localization. (a) Real part and imaginary part of $\Delta S/S$ obtained on the molecular semiconductor DBTTT and of calculations based on the Drude model shown by symbols and solid lines, respectively. (b) Calculations based on the Drude-Smith model, the localization-modified Drude model, and on the Drude-Anderson model, shown by dash-dotted, dashed, and dotted lines, respectively.

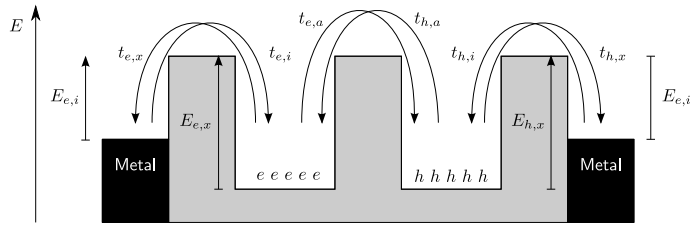


Fig. 6.4: File: zz103001.eps. Transfer processes of trapping model 1.

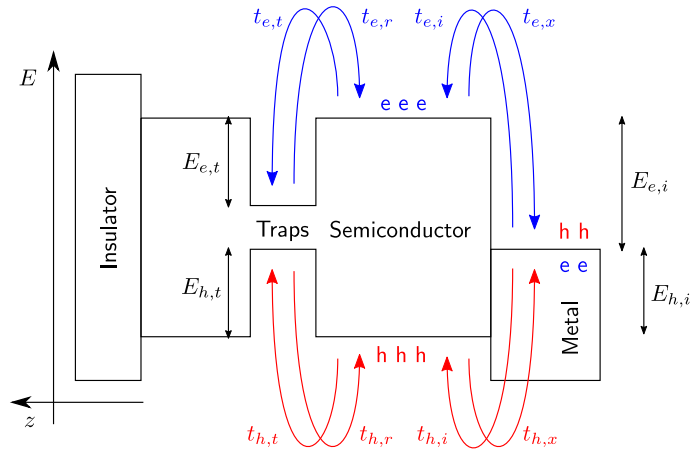


Fig. 6.5: File: zz105001.eps. Transfer processes of trapping model 2.

Fig. zz110: Drude conductivity vs. Lorentz conductivity

Fig. zz114: Drude permittivity

Fig. zz115: Drude conductivity

Fig. zz118: WKB approximation

Fig. zz121: Screening of AC fields

Fig. zz122: Debye length and average distance

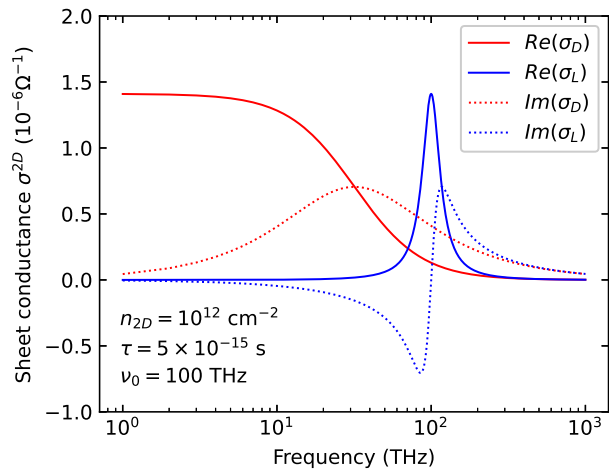


Fig. 6.6: File: zz110001.eps. Drude conductivity vs. Lorentz conductivity

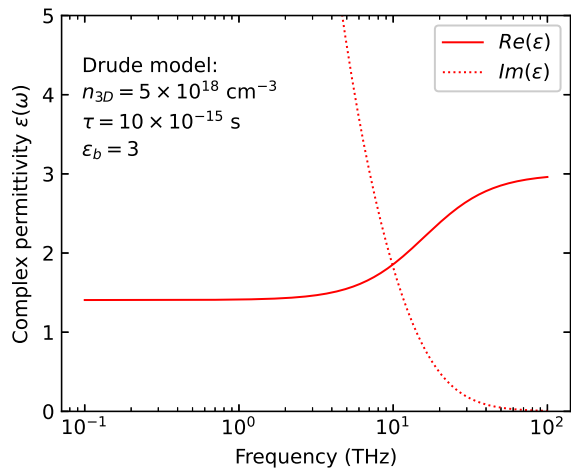


Fig. 6.7: File: zz114001.eps. Drude permittivity

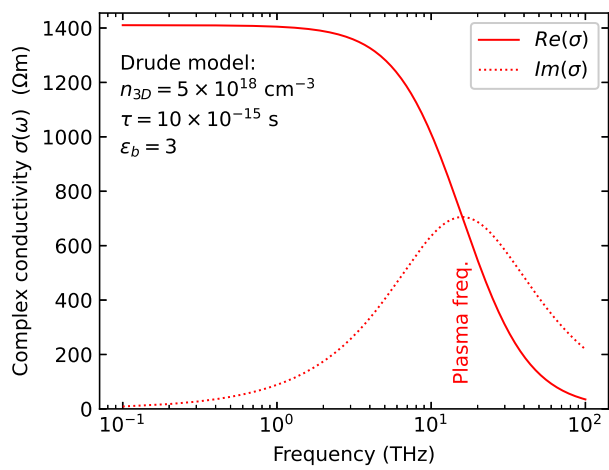


Fig. 6.8: File: zz115001.eps. Drude permittivity

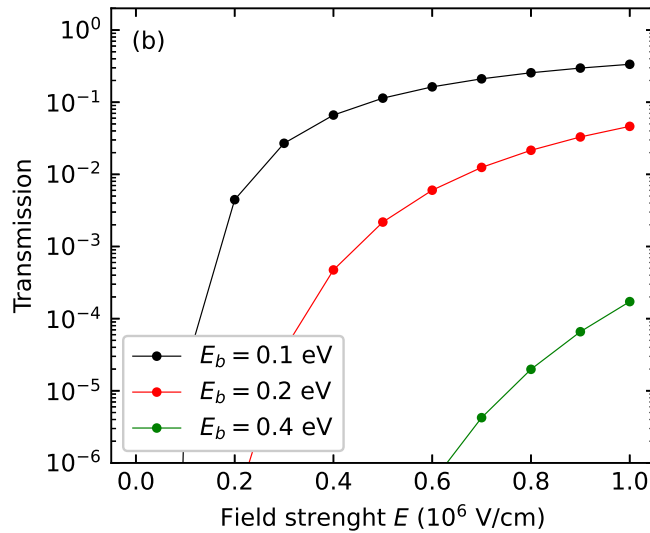
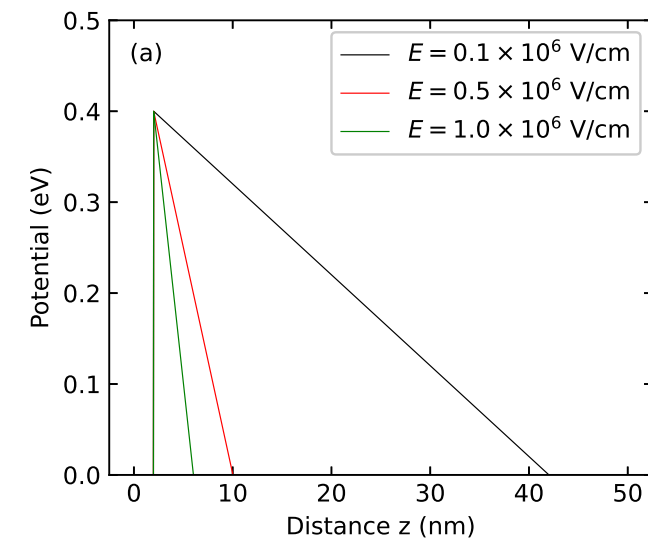


Fig. 6.9: File: zz118001.eps. Tunneling from the injection metal into the semiconductor calculated with WKB approximation.

Fig. zz125: Schematic of plasma oscillations

Fig. zz126: Drude permittivity

Fig. zz133: Determining the effective carrier mass

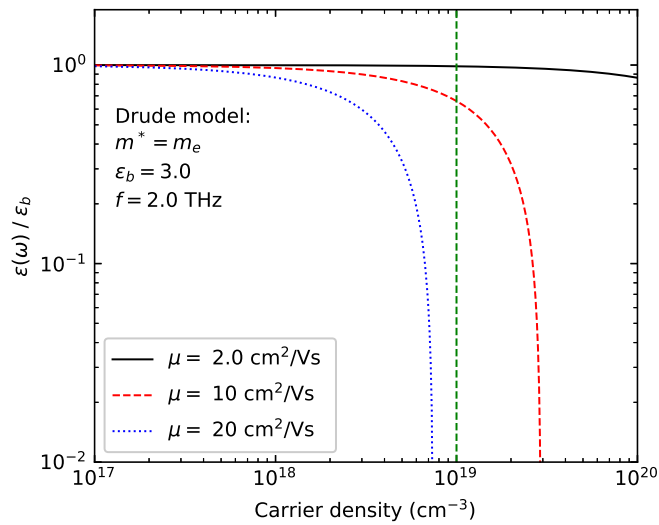


Fig. 6.10: File: zz121001.eps. Screening of AC fields in dependence on the density of mobile carriers.

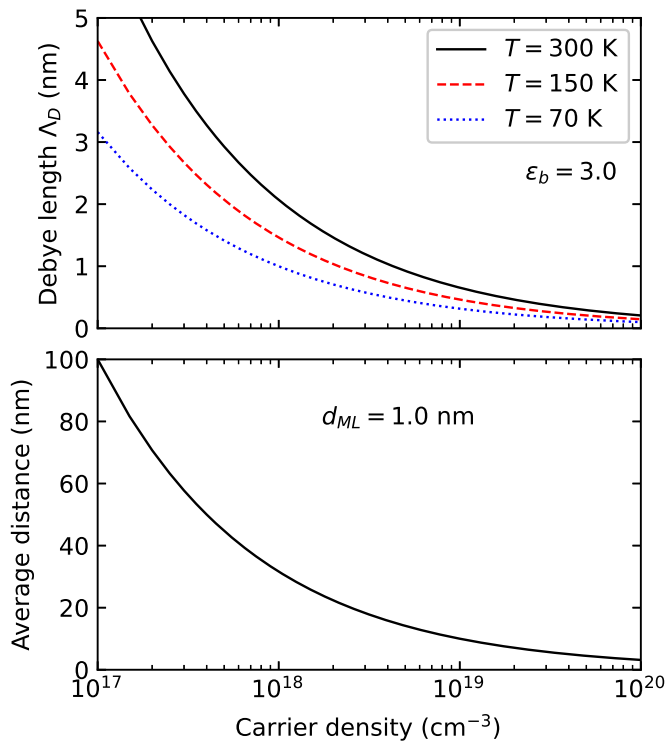


Fig. 6.11: File: zz122001.eps. Top: Debye screening length and average distance. Bottom: Average spacing between carriers.

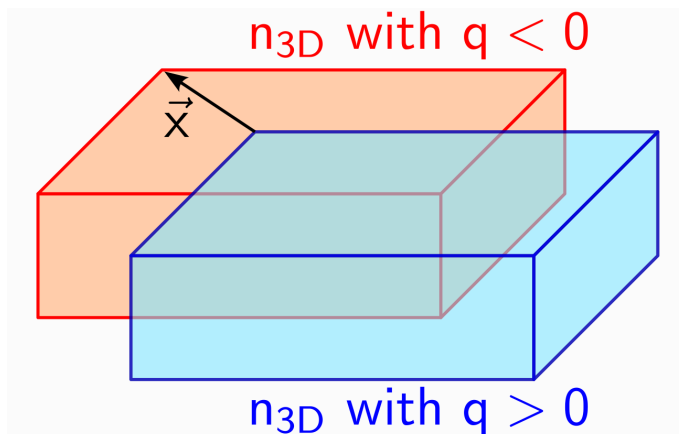


Fig. 6.12: File: zz125001.eps. Schematic of plasma oscillations

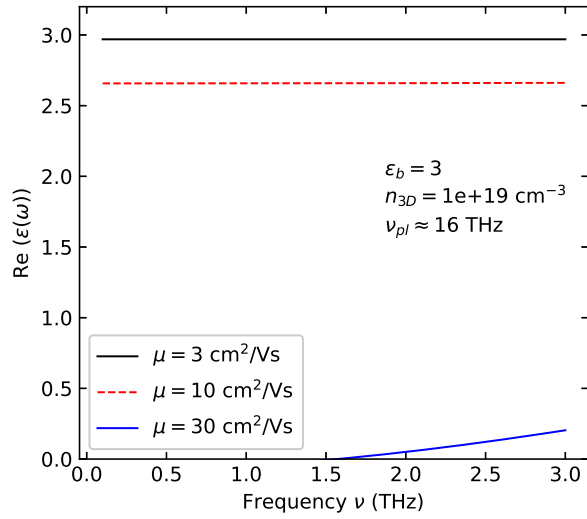


Fig. 6.13: File: zz126001.eps. Real part of the Drude permittivity in the channel of an organic semiconductor.

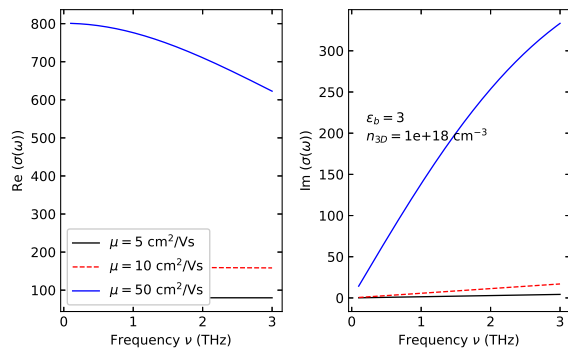


Fig. 6.14: File: zz133001.eps. The effective mass of carriers can be measured when the spectral dependence of the conductivity is no longer linear. The results do not depend on density of the charge carriers or on background permittivity.

7 Simulations

Fig. zz001: Screening Newton project 051

- Data: Newton project 051

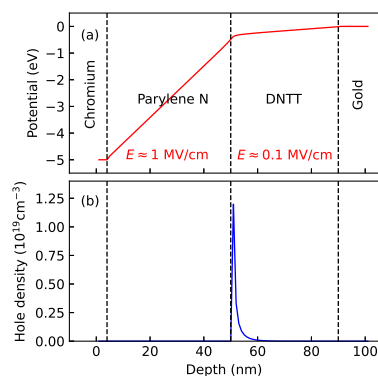


Fig. 7.1: File: zz001001.eps. Screening, Newton project 051.

Fig. zz123: Screening Newton project 070

- Data: Newton project 070

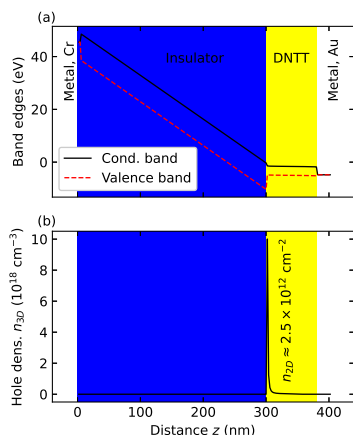


Fig. 7.2: File: zz123001.eps. Screening, Newton project 070.

Fig. zz053: Hole density is first MLs

- Data: Newton project 053
- randomly generated interface, program Rhhc001011.vbp

Fig. zz055: Hole density at ML step

- Data: Newton project 058
- ML step
- the original eps is huge: loaded zz05501.png to Inkscape and saved as zz055001.eps

Fig. zz056: Hole density at crystals with different valence band

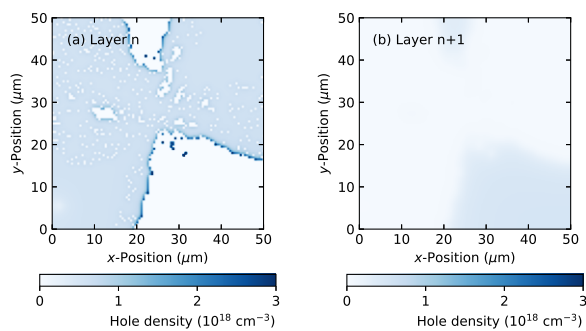


Fig. 7.3: File: zz053001.eps. Hole density in first monolayers. Values are capped to $3 \times 10^{18} \text{ cm}^{-3}$. Newton project 053.

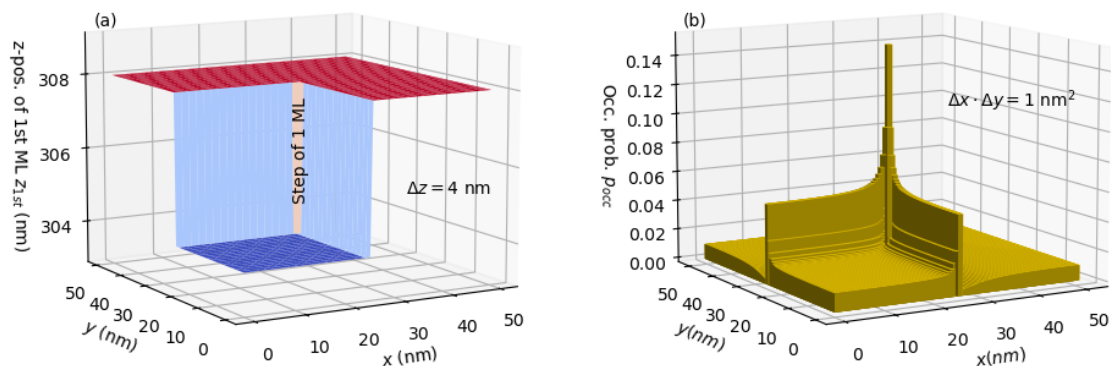


Fig. 7.4: File: zz055001.eps. Newton project 058.

- Data: Newton project 059
- Crystals with different valence band edge

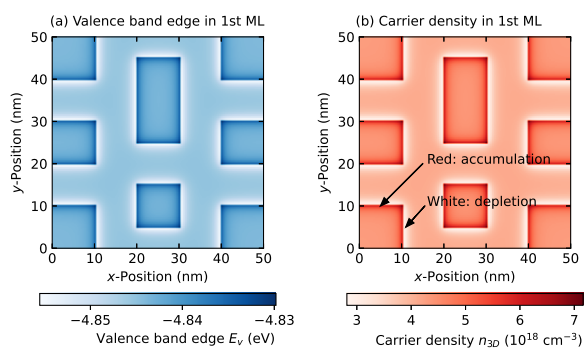


Fig. 7.5: File: zz056001.eps. Newton project 059.

Fig. zz060: Hole density at crystals with different valence band

- Data: Newton projects 060 and 061
- Crystals with different valence band edge

Fig. zz062: Tilted polymer crystals

- Data: Newton project 063

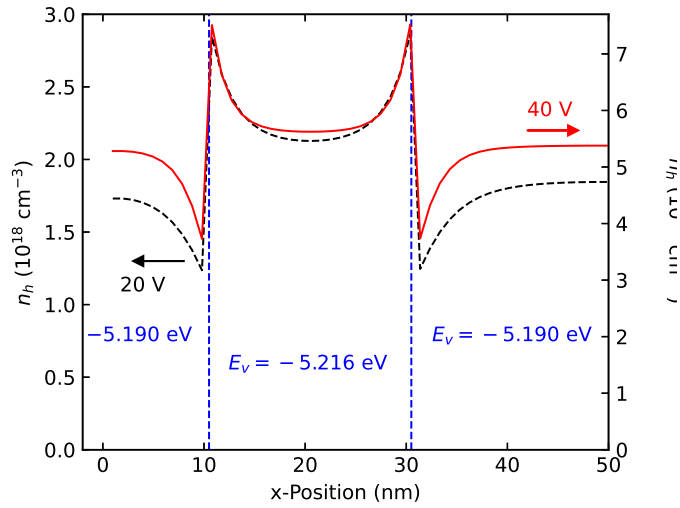


Fig. 7.6: File: zz060002.eps. Newton projects 060 and 061 for voltages of 20 V and 40 V, respectively

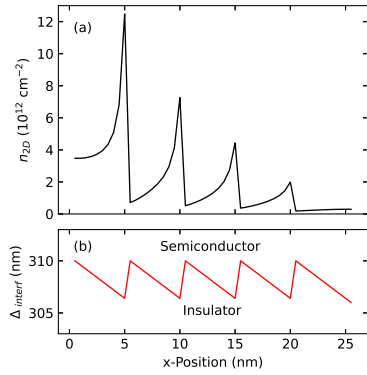


Fig. 7.7: File: zz062001.eps. Tilted polymer crystals, Newton project 063. a) Sheet density in dependence of x-position. The difference peak heights result from the tilted layer surfaces within the cubic volume of the simulation. b) Thickness of the insulator Δ_{insul} between gate contact and semiconductor.

Fig. zz069 and zz070: Tilted polymer layer

- Data: Newton project 068

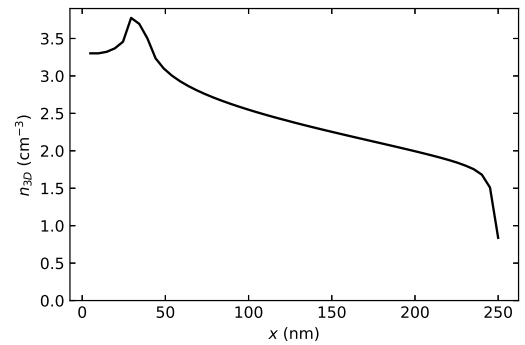
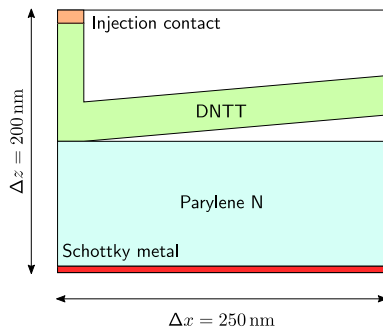


Fig. 7.8: Files: zz069001.eps and zz070001.eps.

Fig. zz072 and zz073: Saw-tooth crystals

- Data: Newton project 069

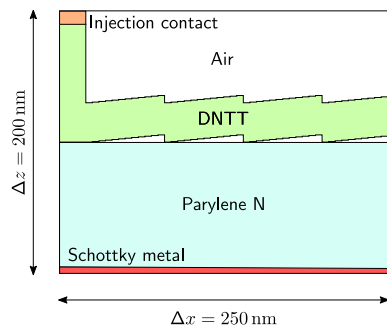


Fig. 7.9: Files: zz072001.eps and zz073001.eps.

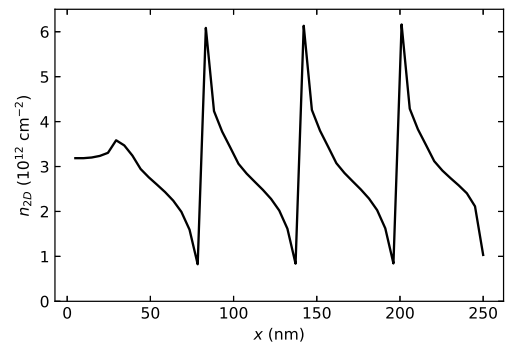


Fig. zz082: Screened defect

- Data: ?

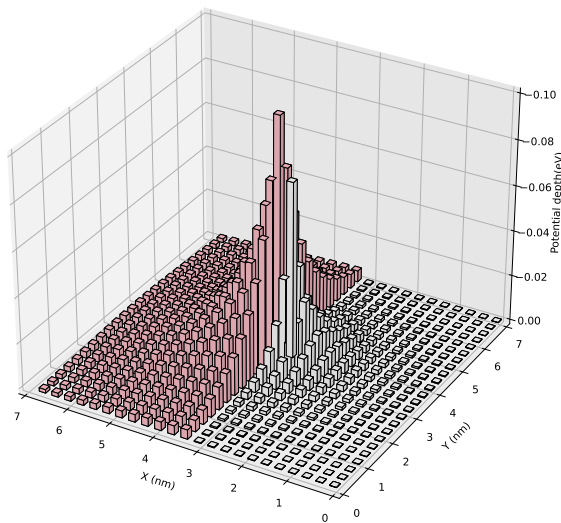


Fig. 7.10: File: zz082001.eps.
Screened defect.

Fig. zz149: Computation time of Newton algorithm

Main result: serious data after 0.5 h.

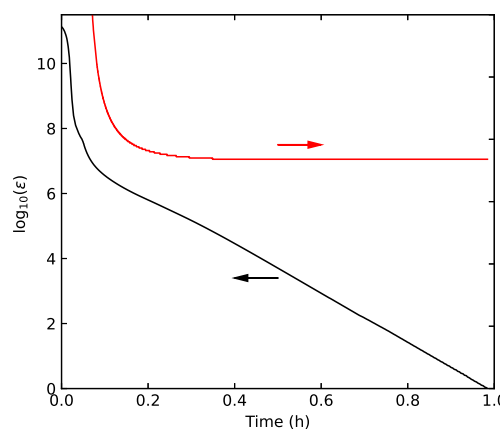


Fig. 7.11: File: zz149. Calculation of a 40*40*100 lattice for van-der-Pauw measurements.

Fig. zz150: Charge density of a gvdP structure as wireframe

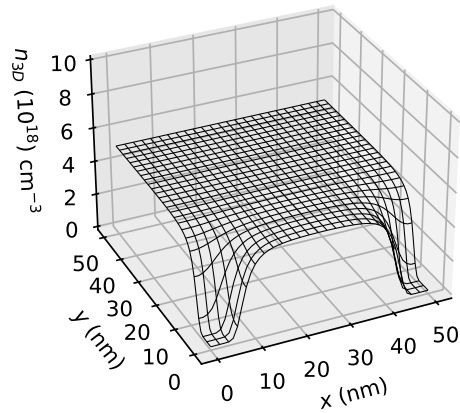
Main results:

- for $V_{34} = 0\text{ V}$ charge depletion around 3 and 4
- for $V_{34} = -4\text{ V}$
 - charge depletion around 4
 - charge accumulation around 3

Details:

- data in ...\\NewtonProject03\\WorkingDir\\Proj002021 to 26
- original graphic in Analyze002021To026
- lattice $51*51*101$

a) $V_{G4} = -5.0\text{ V}$, $V_{34} = 0.0\text{ V}$



b) $V_{G4} = -5.0\text{ V}$, $V_{34} = -4.0\text{ V}$

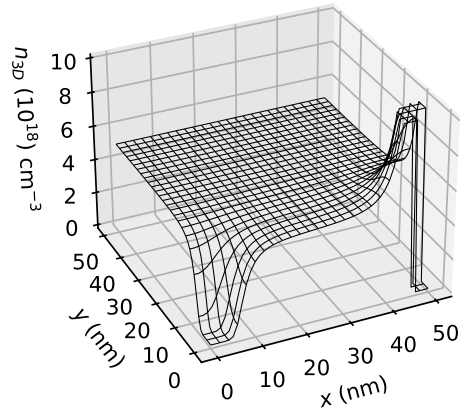


Fig. 7.12: File: zz150. Calculation of the density within a van-der-Pauw structure.

Fig. zz151: Simulated volume

Details:

- data in ...\\NewtonProject03\\WorkingDir\\Proj002021 to 26

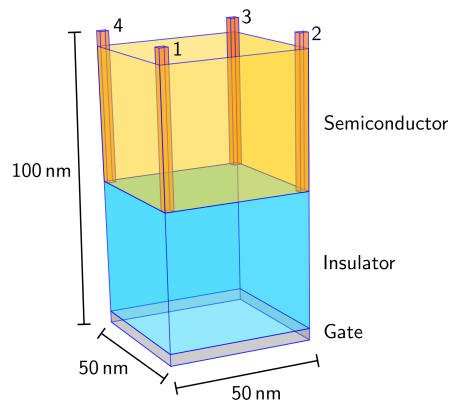


Fig. 7.13: File: zz151. Simulated volume of a van-der-Pauw structure. van-der-Pauw structure.

Fig. zz153: Potential changes between contcts 3 and 4

Details:

- data in ...\\NewtonProject03\\WorkingDir\\Proj002021 to 26
- used data files: see python file

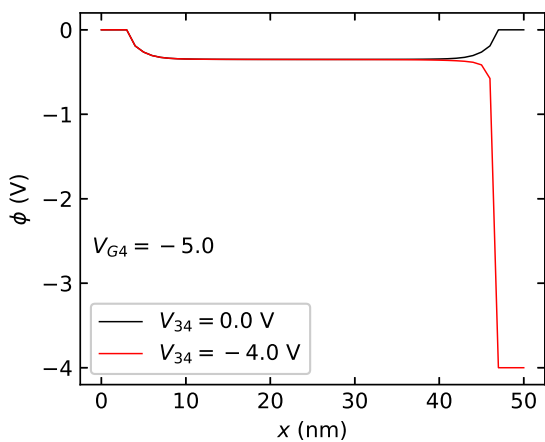


Fig. 7.14: File: zz153. Potential changes between contacts 3 and 4.

Fig. zz154: Steps in insulator/semiconductor interface

Details:

- data in ...\\NewtonProject03\\WorkingDir\\Proj002031 to 36
- used data files: see python file

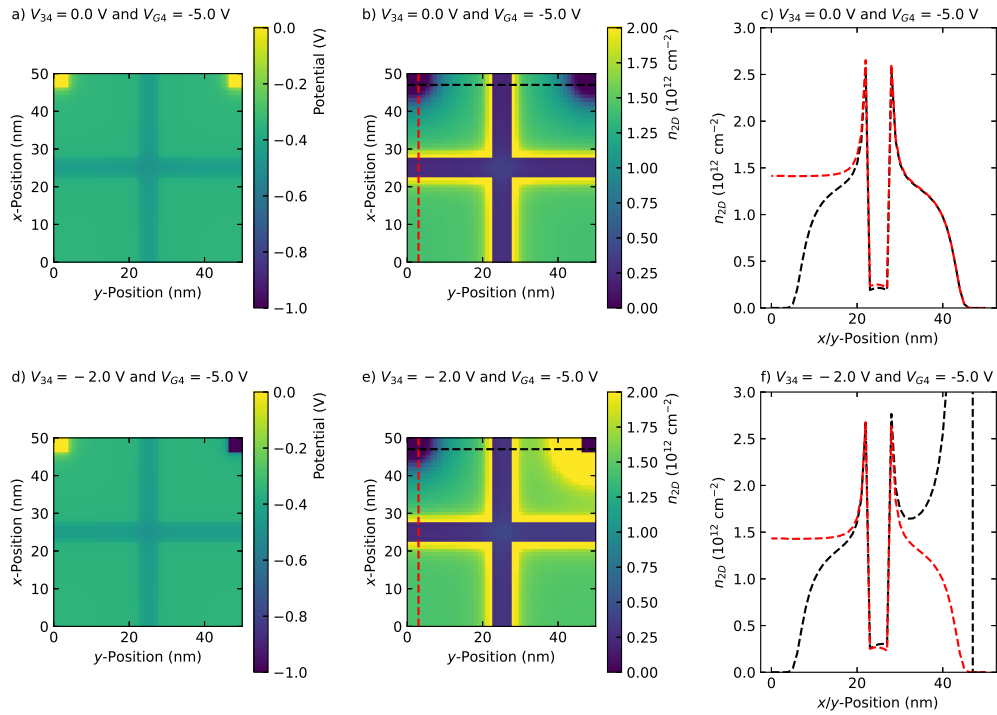


Fig. 7.15: File: zz154. Steps in insulator/semiconductor interface. van-der-Pauw structure.

8 Electrical characterization

Fig. zz026: DLTS pulse sequence

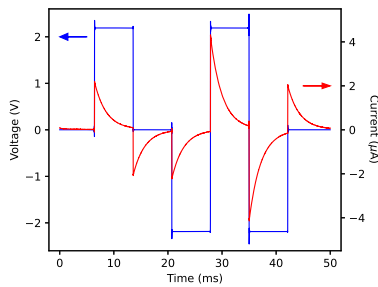


Fig. 8.1: File: zz026001.eps. DLTS measurement on C8-BTBT by TC.

Fig. zz093: DLTS measurements on mica capacitors

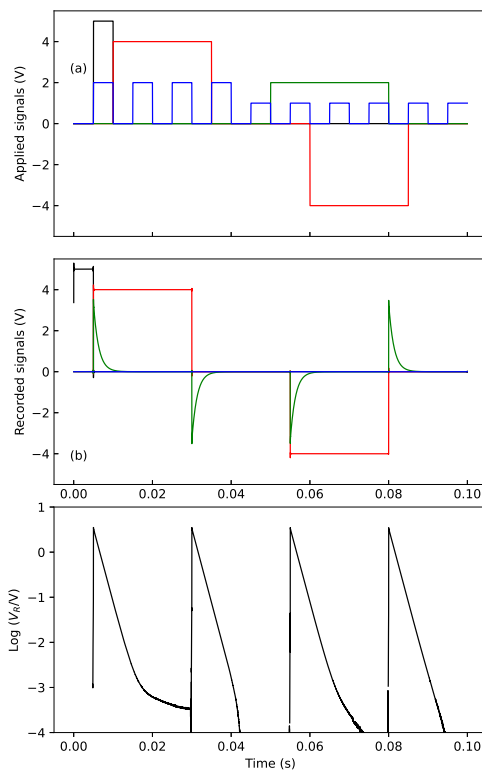


Fig. 8.2: File: zz093002.eps. DLTS measurement on a mica capacitor.

Fig. zz107: Irrelevant device model

Fig. 8.3: File: zz107001.eps. Schematic of device with lateral injection contacts and equivalent capacitors.

Fig. zz093: DLTS measurements on mica capacitors

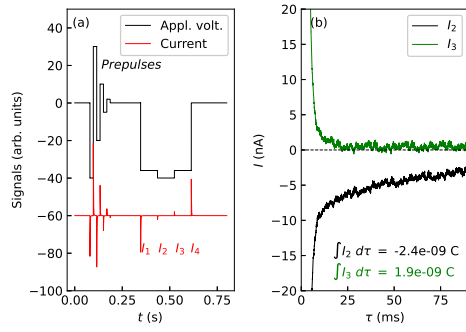


Fig. 8.4: File: zz108001.eps. Detection of trapped carriers by time-resolved deep-level spectroscopy. The pre-pulse sequence removes trapped carriers from the device. The following voltage sequences charges and discharges the device. I_2 and I_3 are nearly differential charging and discharging currents, respectively. From their temporal difference, the density of trapped carriers can be deduced.

Fig. zz136: III-performed DLTS experiment

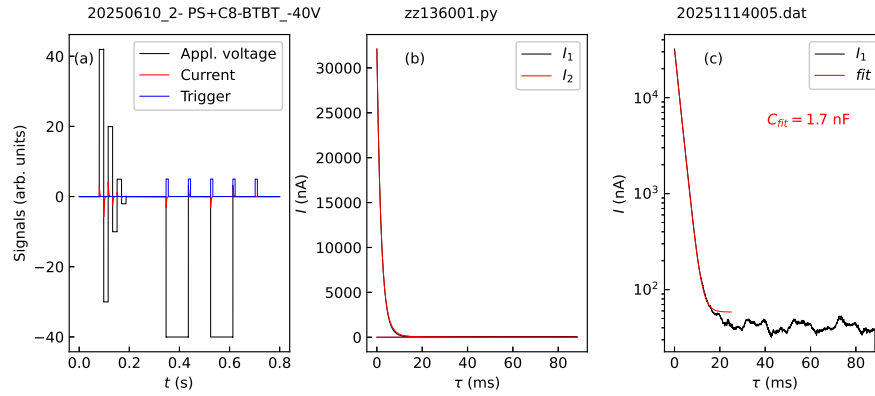


Fig. 8.5: File: zz0136001.eps. DLTS measurement on sample 20250610-2, which is also suited for gvdP. Result: From $C_{fit} = 1.7$ nF, we conclude that charge carriers can be injected sufficiently fast.

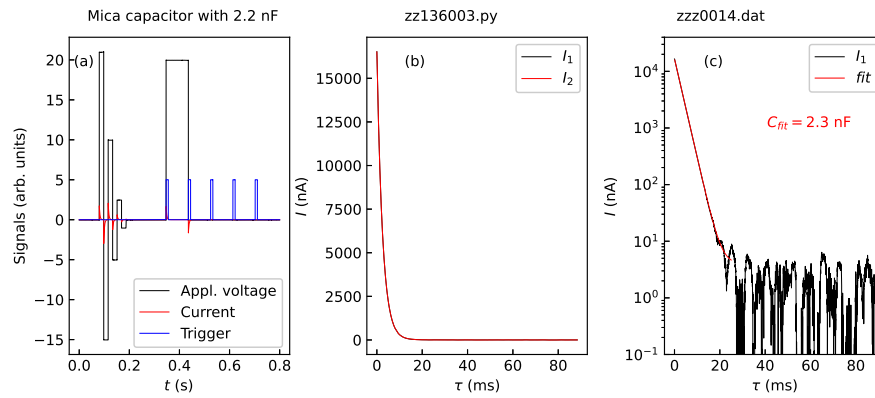


Fig. 8.6: File: zz0136003.eps. DLTS measurement on a mica capacitor with nominally 2.2 nF. Result: Both measurement technique and fitting algorithm work.

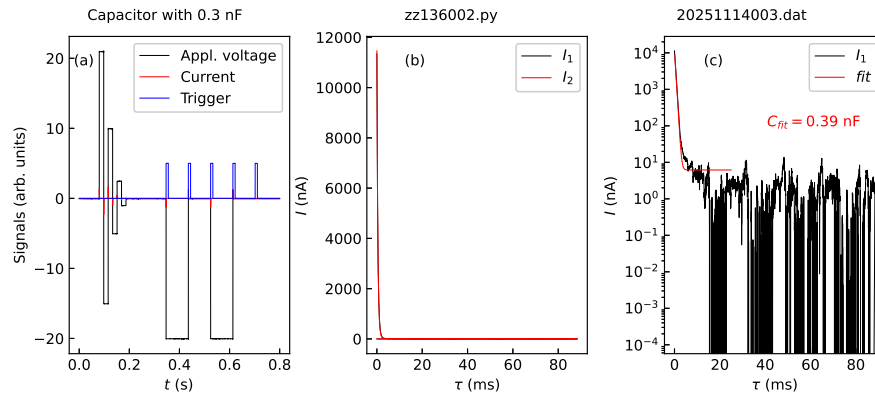


Fig. 8.7: File: zz0136002.eps. DLTS measurement on a capacitor with nominally 0.3 nF. Result: Both measurement technique and fitting algorithm work.

Fig. zz137: Layout for gated van der Pauw experiments

Fig. zz139: Circuit for injecting HF voltages into gvdP experiments

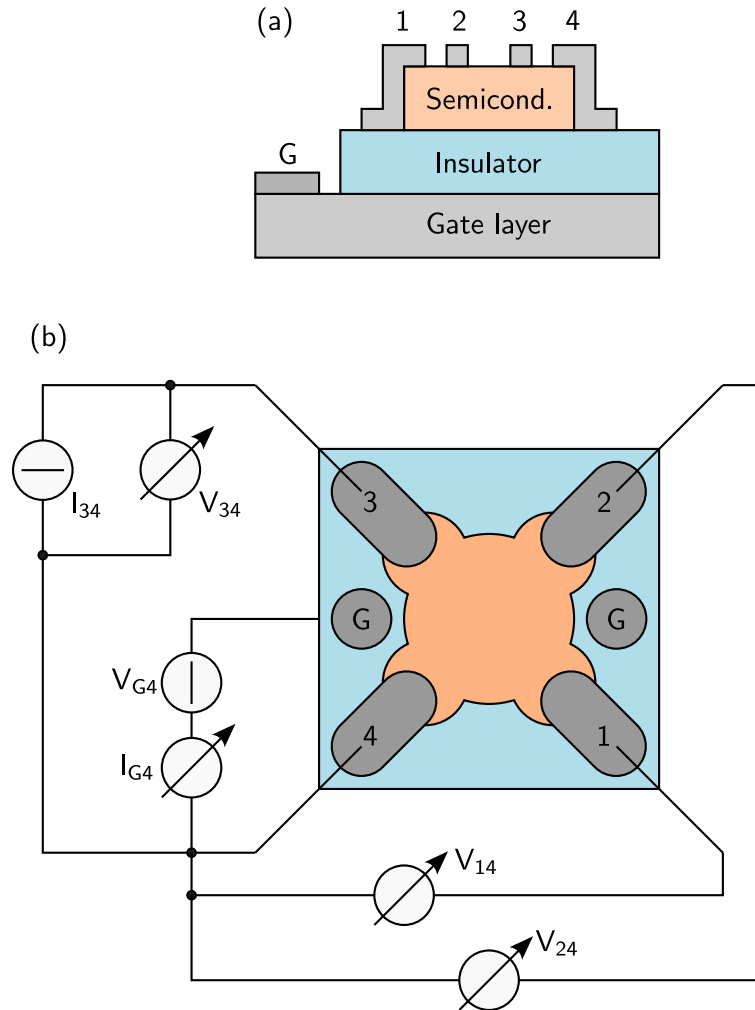


Fig. 8.8: File: zz0137003.eps. Circuit layout for van der Pauw measurements.

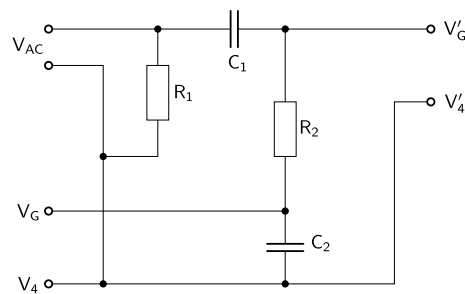


Fig. 8.9: File: zz0139002.eps. Circuit layout for van der Pauw measurements with an additional high-frequency stimulus. The inputs V_{AC} and V_G are decoupled from the output (V'_G and V'_4) by a high-pass and a low-pass, respectively. Recommended values are $R_1 = R_2 = 50 \Omega$, $C_1 = 100 \text{ nF}$, and $C_2 = 10 \mu\text{F}$. This gives critical frequencies of $f_{c1} = 32 \text{ kHz}$ and $f_{c2} = 320 \text{ Hz}$.

Fig. zz142: Readout times of the MM2000

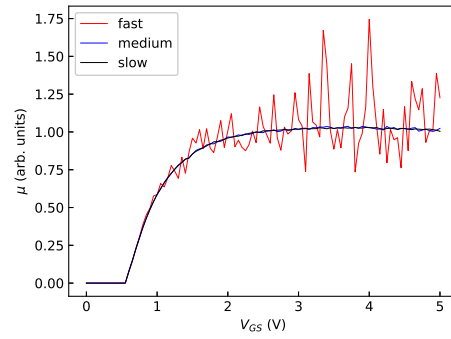


Fig. 8.10: File: zz0142001.eps. Comparisons of the readouts with the MM2000.

Fig. zz143: gvdP measurements on 20251030-9-PS

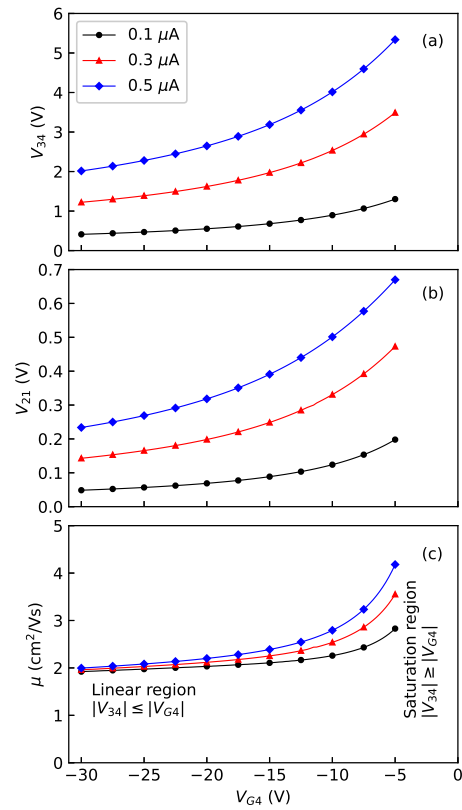


Fig. 8.11: File: zz143001.eps. Van der Pauw measurements on sample 20251030-9-PS.

Fig. zz144: Output characteristics on 20251030-9-PS

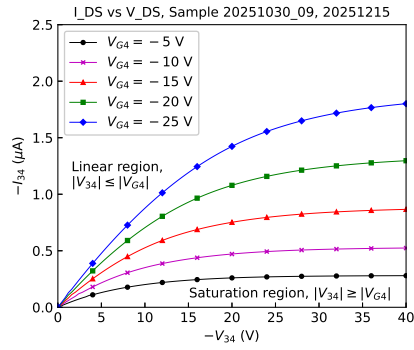


Fig. 8.12: File: zz144.eps. Output characteristics recorded on sample 20251030-9-PS.

Fig. zz145: Transfer characteristics on 20251030-9-PS

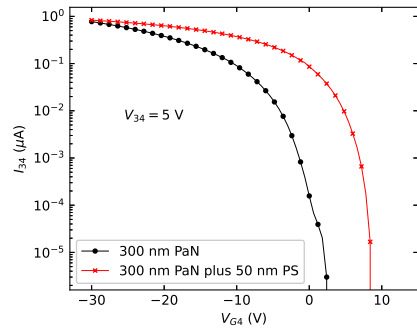


Fig. 8.13: File: zz145.eps. Transfer characteristic on sample 20251030-9-PS (???)

Fig. zz152: Constant background resistances from gvdP on 20251030-9-PS

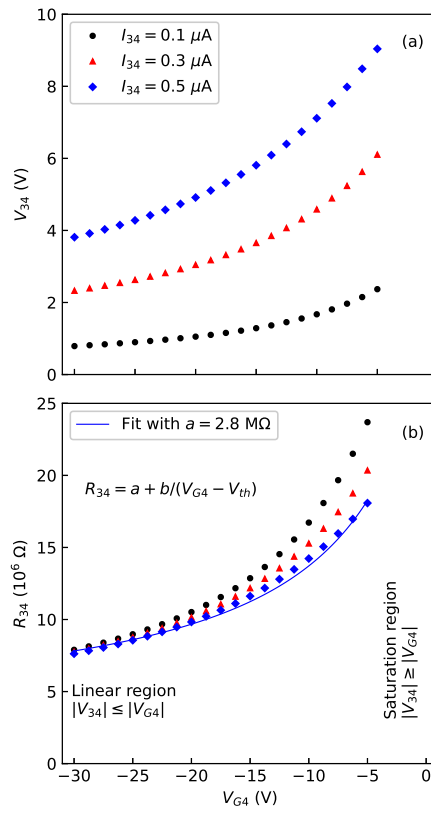


Fig. 8.14: File: zz152.eps. Extracting a constant resistance on vdP data recorded on sample 20251030-9-PS (???)

9 Optical characterization

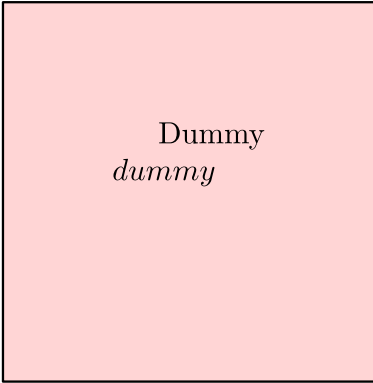


Fig. 9.1: File: zz000000.eps. Dummy.

10 AFM characterization

Fig. zz027001:

- Description:
 - AFM
 - 2*2mu, 512*512
 - by: P.R.
- Sample:
 - sample number: 20210302-04
 - C8-BTBT, 30 - 50 nm
 - on PaN, 300 nm
- Original data:
 - Folder: Q:\THz-Lab\Measurement Data\AllMeasuredData\
 - File: 202120210302-4_C8_2x2_4umPs_512.txt

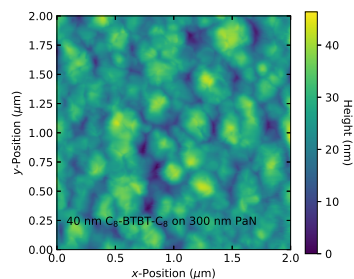


Fig. 10.1: File: zz027001.eps. C8-BTBT on PaN.

Fig. zz028001:

- Description:
 - AFM
 - 2*2mu, 512*512
 - by: P.R.
- Sample:
 - sample number: 20220926
 - C8-BTBT, 30 - 50 nm
 - on PS, approx 70 nm
 - on PaN, approx 300 nm
- Original data:
 - Folder: Q:\THz-Lab\Measurement Data\AllMeasuredData\2022\20221206
 - File: 20220926_C8 on PS_annealed_5x5_6umPs_02.txt

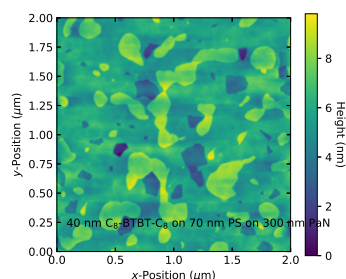


Fig. 10.2: File: zz028001.eps. C8-BTBT on PS on PaN.

Fig. zz029001:

- Description:
 - AFM
 - 2*2mu, 512*512
 - by: P.R.
- Sample:
 - sample number: 20221108-03
 - C8-BTBT, thickness unknown
 - on Trip-C12:
 - mass filled: 15 - 20 mg
 - substrate temp: RT
 - density 1.3 g/cm³
 - soaks 100C and 200C for 100 secs evaporated thickness 5 - 7 nm
 - on PS (approx 75 nm):
 - 2wt
 - 5s with 500 rpm
 - 50 s with 3000 rpm
 - 60 min at 140 C
 - thickness 50 – 100 nm
 - on PaN, 300 nm
- Original data:
 - Folder: Q:\THz-Lab\Measurement Data\AllMeasuredData\2023\20230119\AFM C8 on Trip
 - File: 20221108-03_c8 on trip_5x5.txt

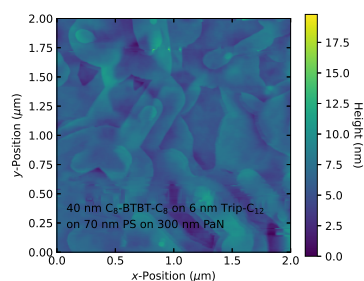


Fig. 10.3: File: zz029001.eps. C8-BTBT on ZTrip-C12 on PS on PaN.

Fig. zz066:

- Description:
 - AFM
 - 5*5mu
 - by: J.S.
 - original data: 20240612001.spm
- Sample:
 - Polyethylene thin-film

Fig. zz074:

- AFM resolution
- by: J.S.
- Topic: P3HT, AFM image, data from 2024062403.spm
- Phase data -j zz074004.dat

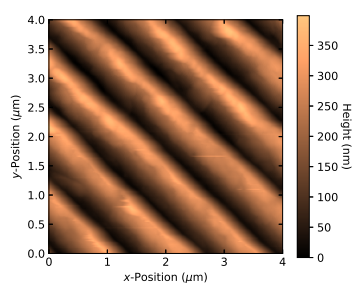


Fig. 10.4: File: zz066001.eps. Impring with a 1200 lines/mm grating on polyethylene

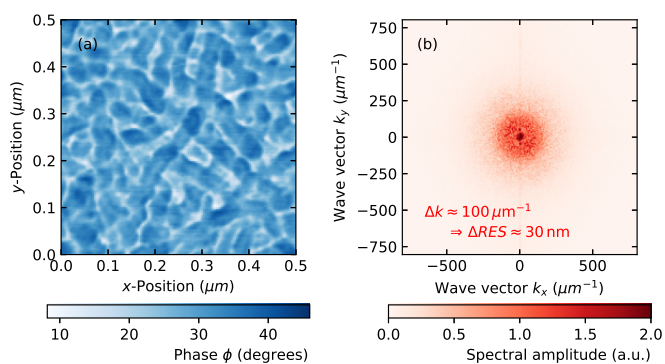


Fig. 10.5: File: zz074001.eps. AFM phase data recorded on P3Ht after 4 h acetone.

11 Device technology

Fig. zz080: Foto of devices on Teonex

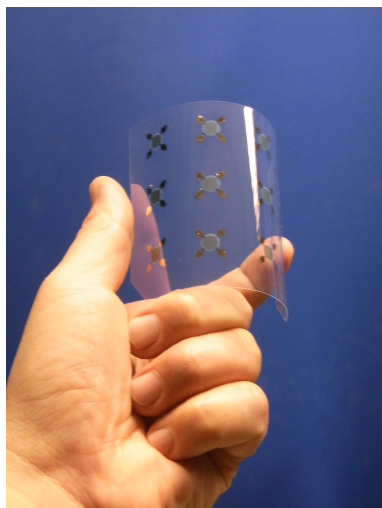


Fig. 11.1: File: zz080001small.eps. Devices on Teonex.

Fig. zz081: Layout of devices

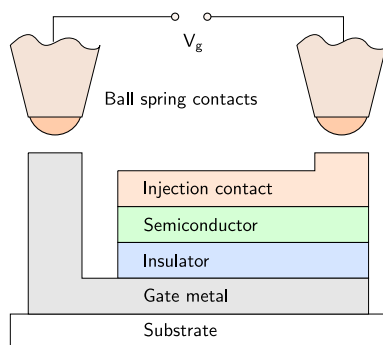


Fig. 11.2: File: zz081001.eps. Layout of devices.

Fig. zz087: Alignment of P3HT nanofibers in imprint

Original data:

- 20240702003.spm: P3Ht thin film on grating
- 20240606001.spm: PE grating

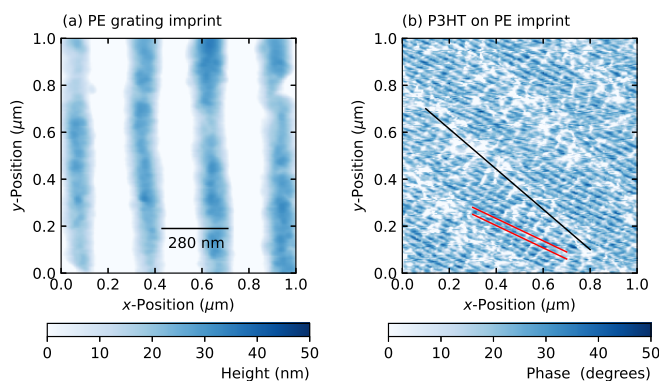


Fig. 11.3: File: zz087001.eps. (a) AFM image of an imprint by a 280 nm-grating into PE. (b) P3HT thin-film on the 280 nm-grating fabricated by off-center spin casting. The red lines indicate the direction of the P3HT nanofibers. The black line indicates the direction of the grating lines. The slight misalignment between grating lines and nanofibers may result from the local centrifugal forces during spin-coating.

Fig. zz095: Possible device layouts

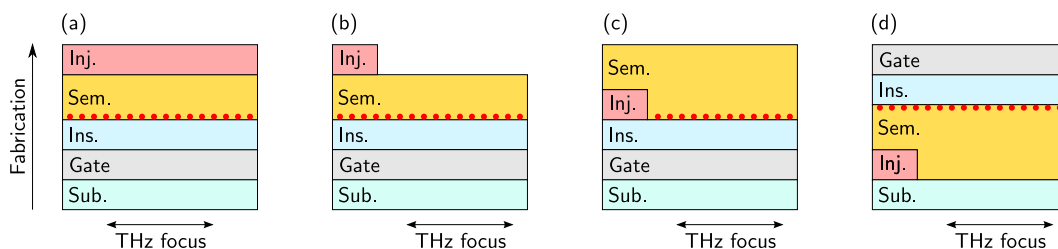


Fig. 11.4: File: zz095001.eps. Possible device layouts using Sub. for substrate, Gate for gate metal, Ins. for insulator, Sem. for semiconducting polymer, and Inj. for injection contact. Accumulated charge carriers are illustrated by red dots. (a) Layout used in all previous THz studies. (b) Similar to bottom-gate top contact structures of FETs. (c) Bottom-gate top-contact. (d) Inverted structure with top-gate bottom-contact geometry.

Fig. zz098: Differential detection of trap density

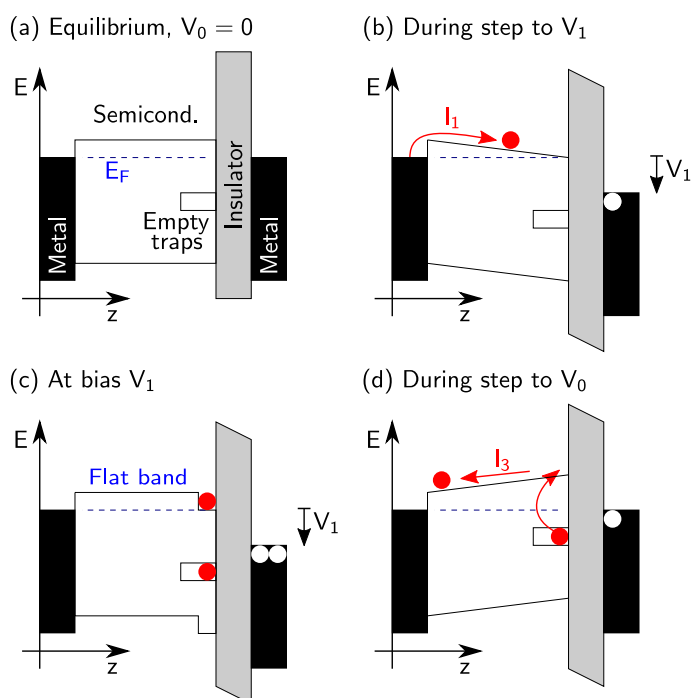


Fig. 11.5: File: zz098021a.eps. Schematic of bias sequence for the differential detection of traps.

Fig. zz104: Illustration for ads

Fig. zz119: Performace of CVD microbalance

Fig. zz120: Varactor structure

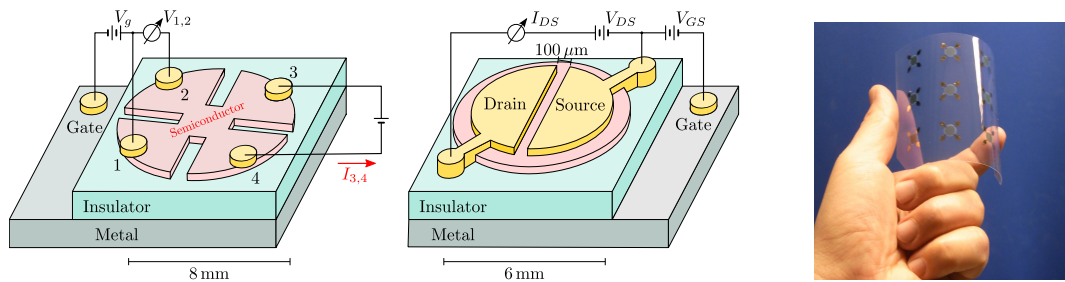


Fig. 11.6: File: zz104001.eps. Illustration for ads.

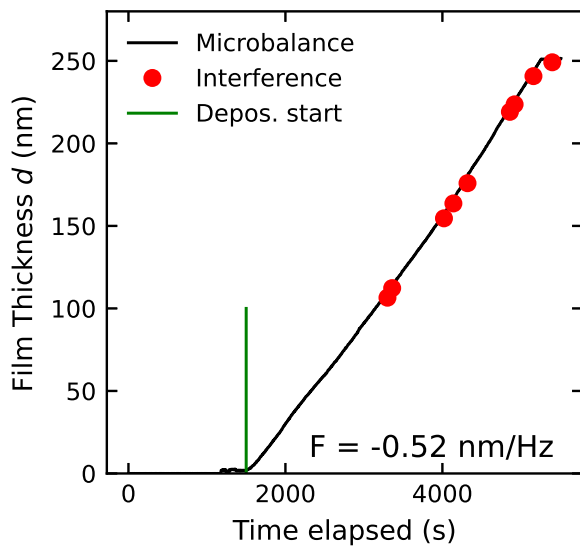


Fig. 11.7: File: zz0119001.eps. Deposition of parylene N. Film thickness measured with microbalance and by optical inspection of interference colors.

Fig. zz131: Device structure A

Fig. zz132: Device structure B

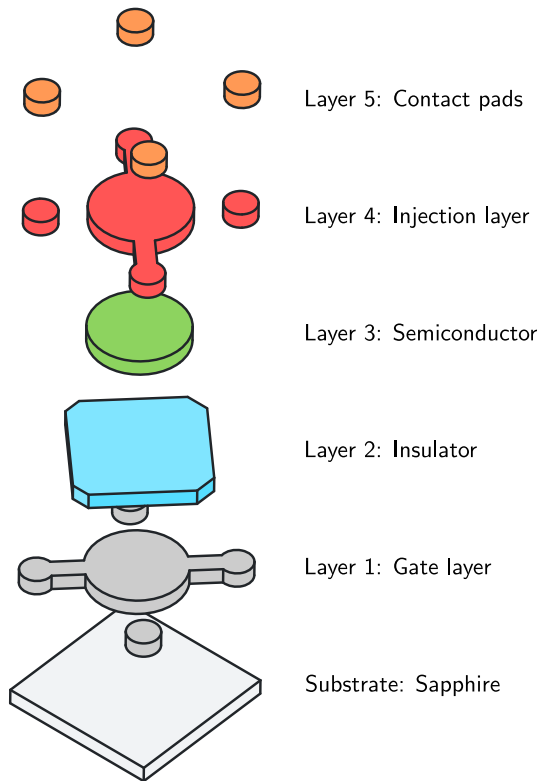


Fig. 11.8: Varactor-type device for THz electromodulation experiments. File: zz0120002.eps. Exploded-view drawing of device structure. Freecad & svg in folder zz120.

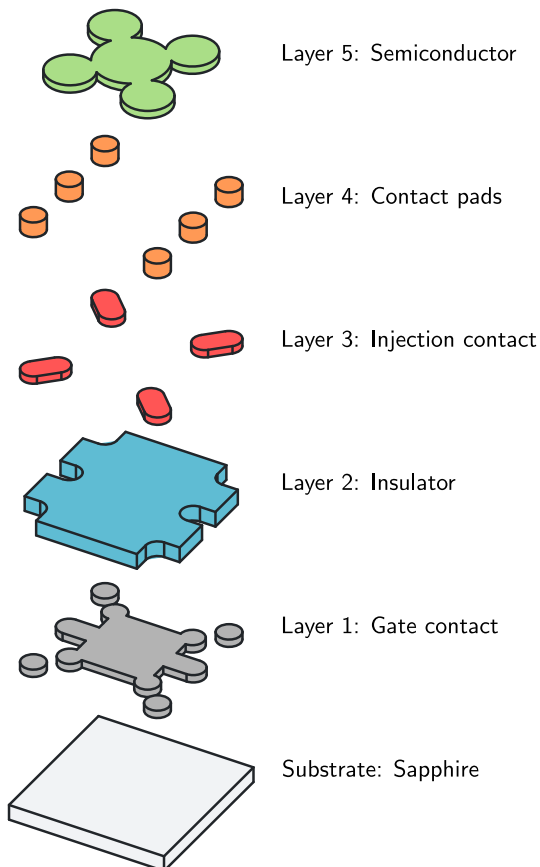


Fig. 11.9: Device architecture A for THz electromodulation experiments and for van der Pauw measurements. File: zz0131002.eps. Exploded-view drawing of device structure. Freecad & svg in folder zz131.

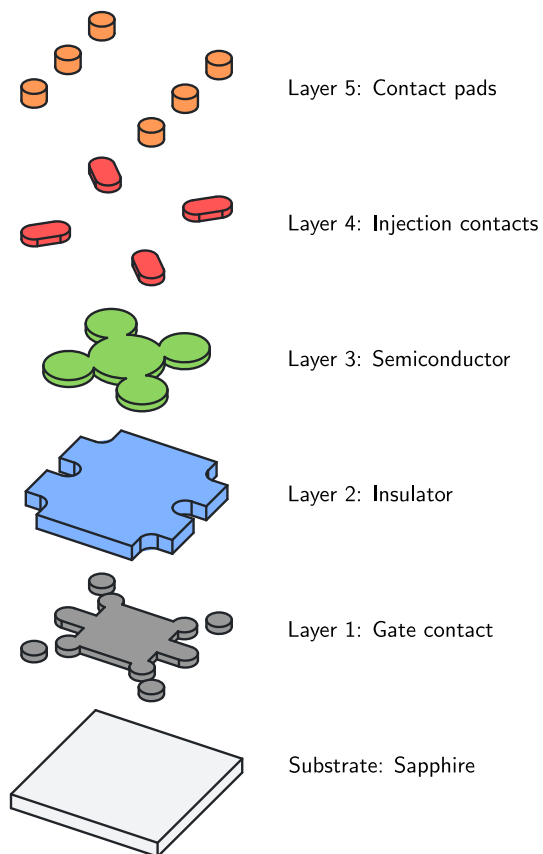


Fig. 11.10: Device architecture B for THz electromodulation experiments and for van der Pauw measurements. File: zz0132001.eps. Exploded-view drawing of device structure. Freecad & svg in folder zz132.

12 Physics

Fig. zz089: Illustration of thermal mean free path ℓ_{th}

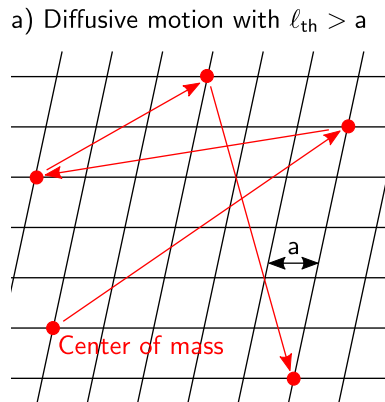


Fig. 12.1: File: zz089001.eps. Illustration of thermal mean free path ℓ_{th}

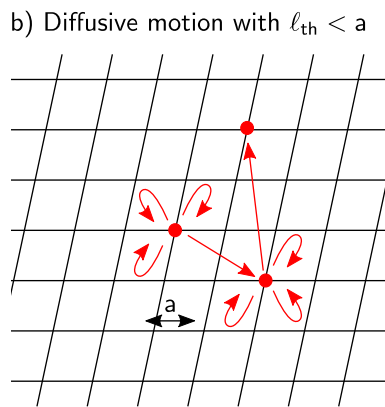


Fig. zz112: Band transport vs. hopping transport

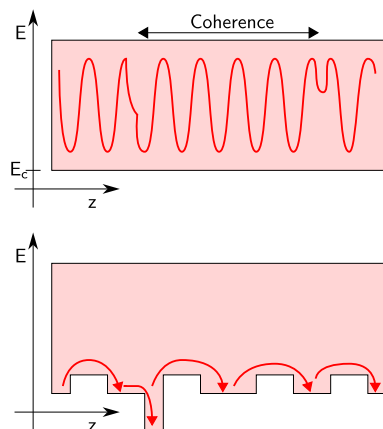


Fig. 12.2: File: zz112001.eps. Illustration of band transport and hopping transport.

Fig. zz113: Momentum relaxation

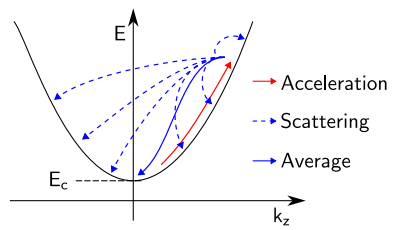


Fig. 12.3: File: zz113001.eps. Momentum relaxation.

13 Electronics

Fig. zz100: Transmission of an LR high pass

Comment: calculated with zz100002.py

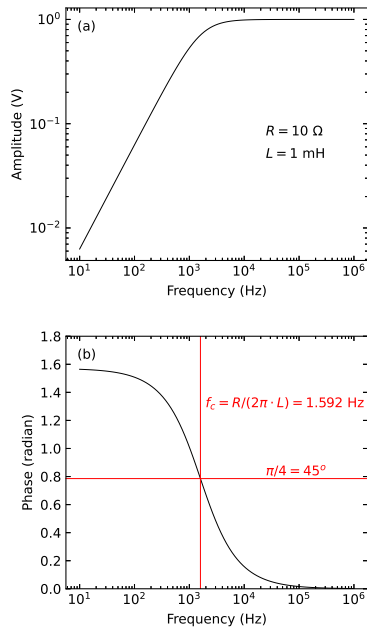


Fig. 13.1: File: zz100001.eps. Transmission of an LR high pass.

Fig. zz102: Transmission of an RC low-pass

Comment: calculated with zz102001.py

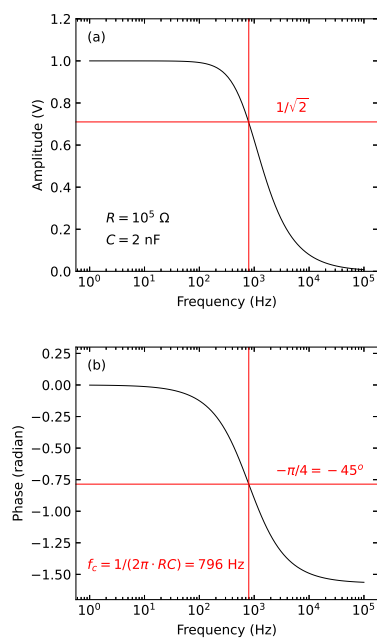


Fig. 13.2: File: zz102001.eps. Transmission of an RC low-pass.

14 Tests of computation

Fig. zz134: Handling of FFT with pythn

Comment: calculated with zz134001.py

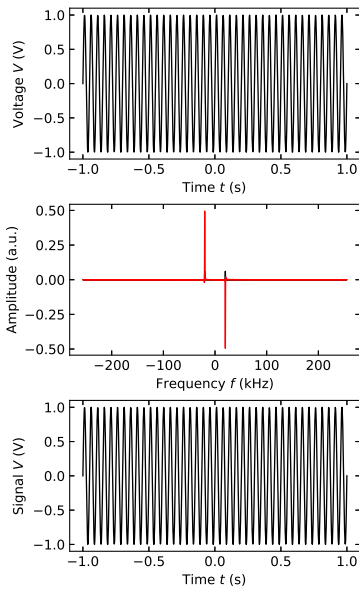


Fig. 14.1: File: zz134001.eps. Test of FFT in python.

15 References

3 Bibliography

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